Genome Wide Association Studies
– GWAS –

Dr. Cécile Ben; Prof. Laurent Gentzbittel

École Nationale Supérieure Agronomique de Toulouse – EcoLab

International Course
Plant Quantitative Genetics, GWAS & Genomic Prediction

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What is association genetics ?

General meaning :

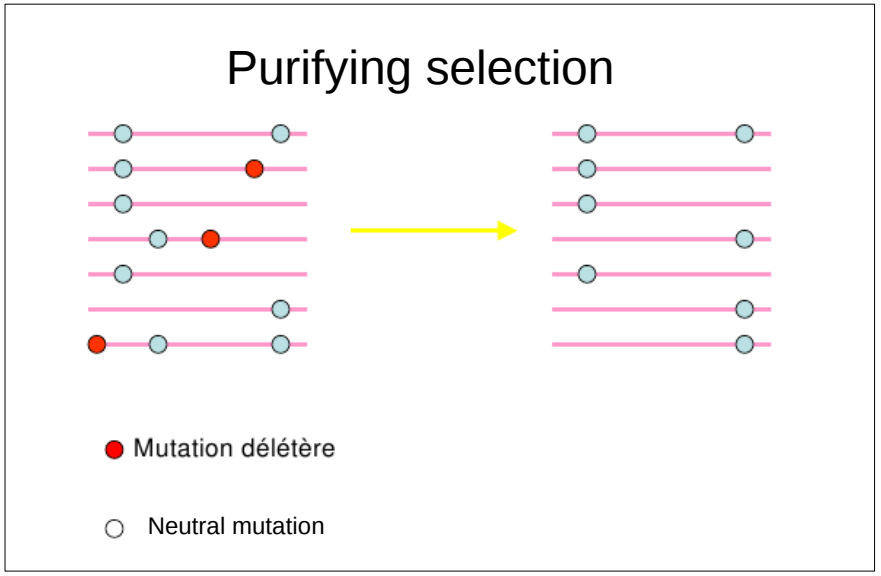
Using natural genetic variability to identify loci of interest.

In the pratice :

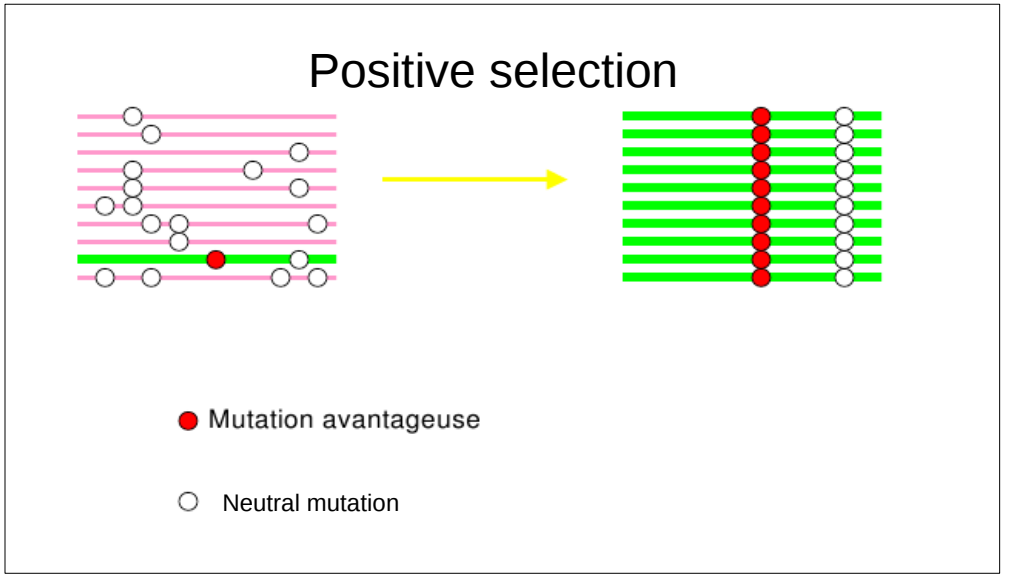
Evaluate phenotypes in a collection of genotypes

for which we have a large number of molecular markers

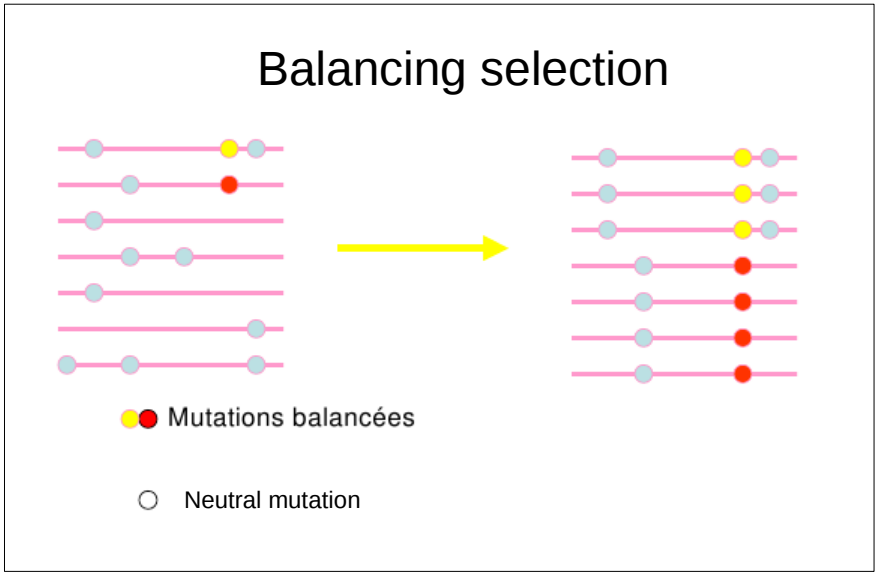
Natural selection modifies allelic frequencies of some genes



50-75 % of polymorphisms in *M. truncatula*



1 % of loci in *M. truncatula*



What will be seen when comparing contrasted lines ?

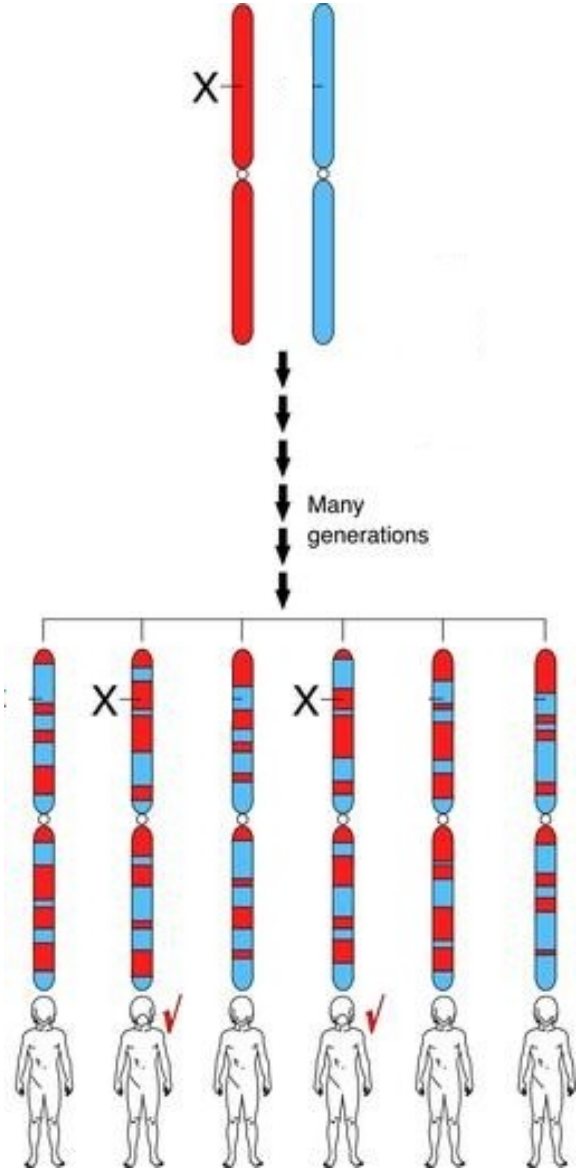
What is linkage disequilibrium ?

Linkage disequilibrium is defined as the difference between the observed frequency of a particular combination of alleles at two loci and the frequency expected for random association.

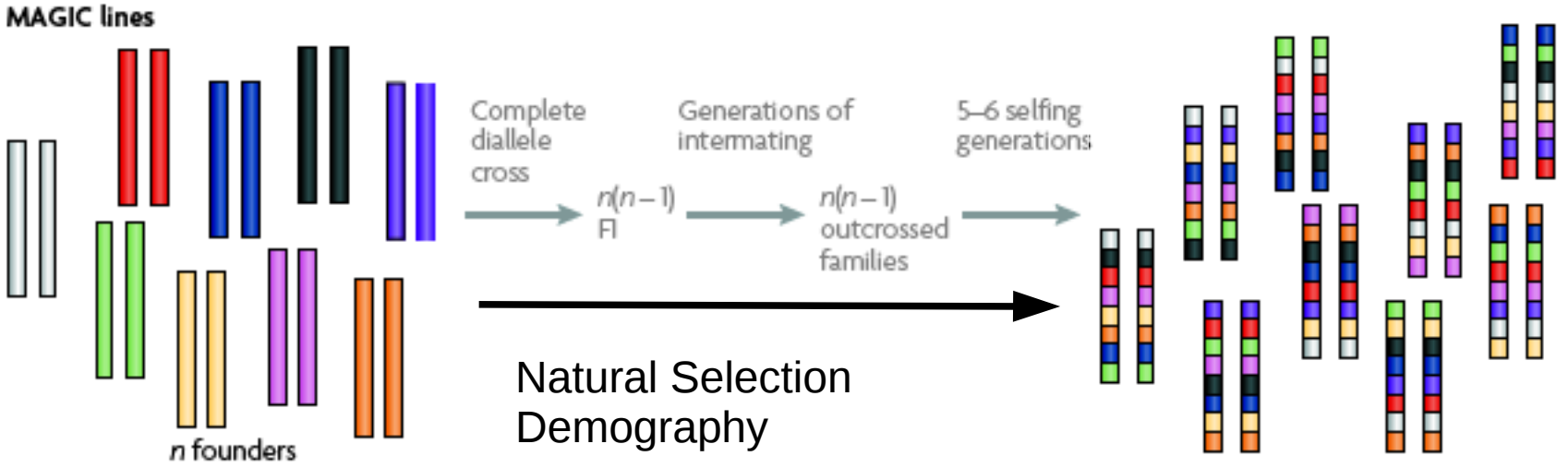
This is based on the assumption that, given sufficient evolutionary time, the occurrence of random recombination events will result in an equilibrium distribution of alleles at each locus.

Thus, the frequency of a particular allele at a given locus will be independent of alleles at other linked loci.

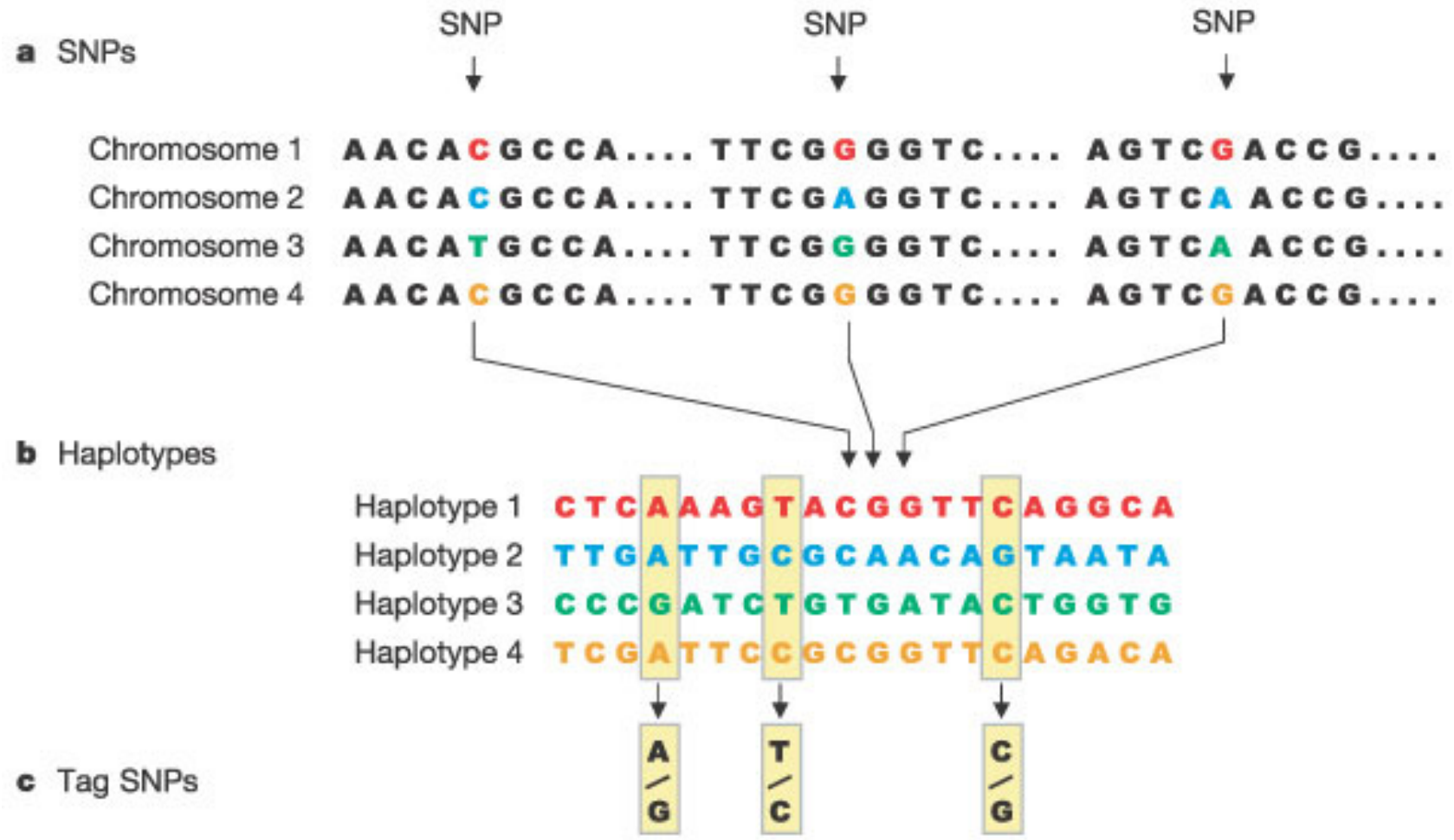
Biological bases of linkage disequilibrium



Biological bases : a gigantic MAGIC



Sequences and haplotypes



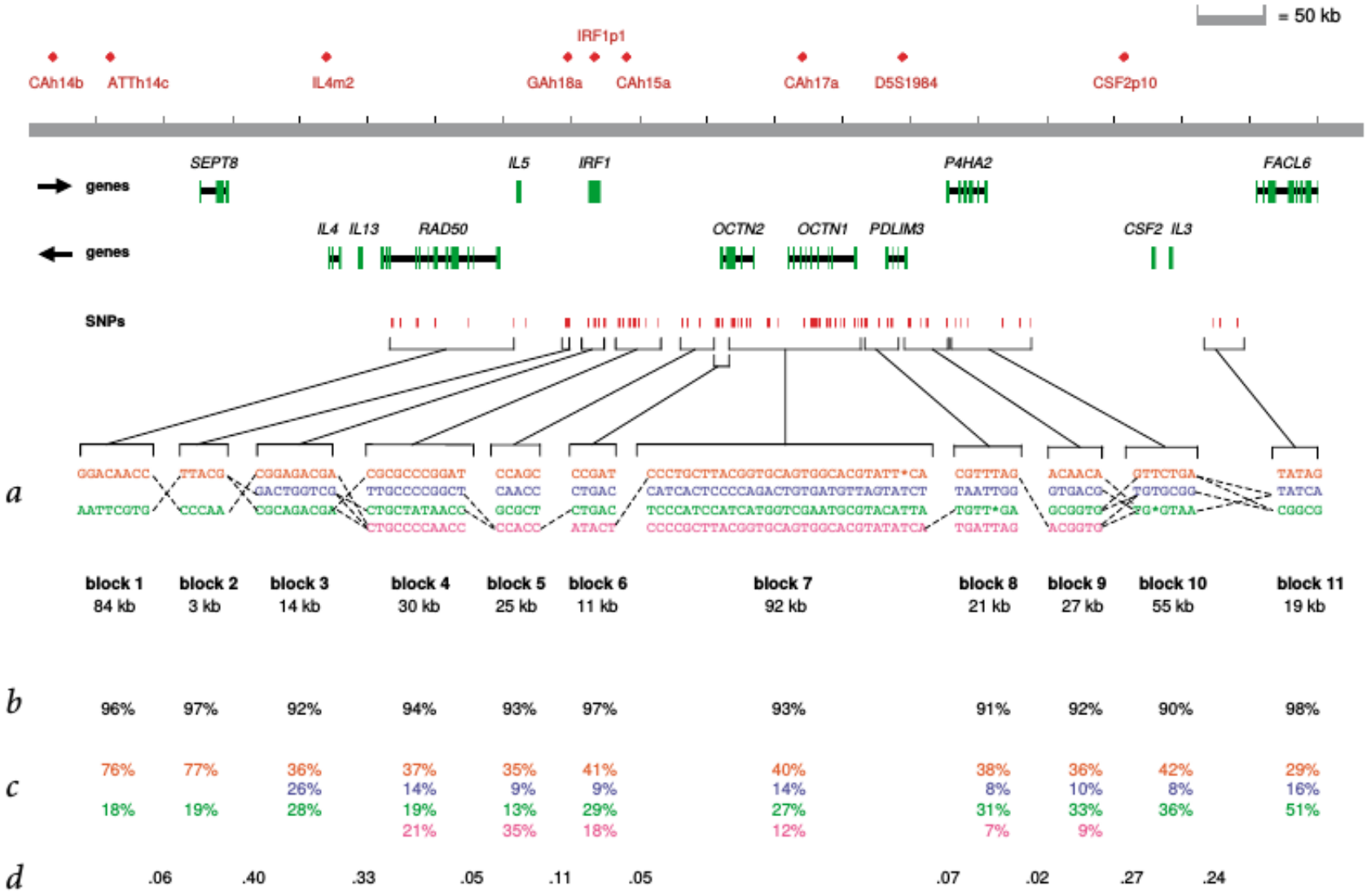
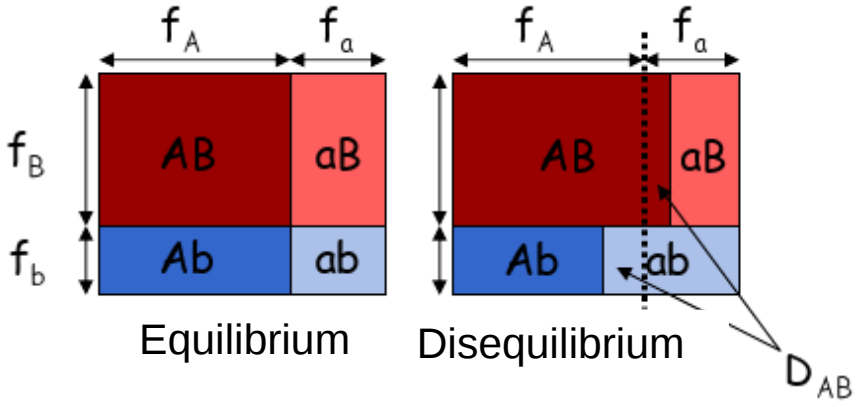


Fig. 2 Block-like haplotype diversity at 5q31. ***a***, Common haplotype patterns in each block of low diversity. Dashed lines indicate locations where more than 2% of all chromosomes are observed to transition from one common haplotype to a different one. ***b***, Percentage of observed chromosomes that match one of the common patterns exactly. ***c***, Percentage of each of the common patterns among untransmitted chromosomes. ***d***, Rate of haplotype exchange between the blocks as estimated by the HMM. We excluded several markers at each end of the map as they provided evidence that the blocks did not continue but were not adequate to build a first or last block. In addition, four markers fell between blocks, which suggests that the recombinational clustering may not take place at a specific base-pair position, but rather in small regions.

Daly, M. J., Rioux, J. D., Schaffner, S. F., Hudson, T. J., and Lander, E. S. (2001). High-resolution haplotype structure in the human genome. Nature Genetics 29, 229–232. doi:10.1038/ng1001-229.

Measures for LD



Equilibrium Disequilibrium

Maximum disequilibrium

$$Dmax = \min(f_A.f_b ; f_B.f_a)$$

(si $f_A > f_a$ et $f_B > f_b$)

Normalised disequilibrium

$$D' = D/Dmax$$

Lewontin, 1964

Linkage equilibrium : $f_{AB} = f_A.f_B$

Disequilibrium : $D_{AB} = f_{AB} - f_A.f_B$

$= f_{AB}.f_{ab} - f_{aB}.f_{Ab}$

$= D_{ab} = -D_{Ab} = -D_{aB}$

$f_{AB} = f_A.f_B + D_{AB}$

$f_{Ab} = f_A.f_b - D_{AB}$

$f_{aB} = f_a.f_B - D_{AB}$

$f_{ab} = f_a.f_b + D_{AB}$

Correlation between sites

$$r^2 = D^2 / (f_A.f_a.f_B.f_b) = \rho^2$$

Hill et Robertson, 1968

Multi allelic LD

Multi locus LD

Examples of LD computations



Site2 \ Site1	G	A
T	6	0
C	0	3

|D|=0.22; Dmax=0.22
|D'|=1; r^2=1

Site3 \ Site1	G	A
G	2	1
A	4	2

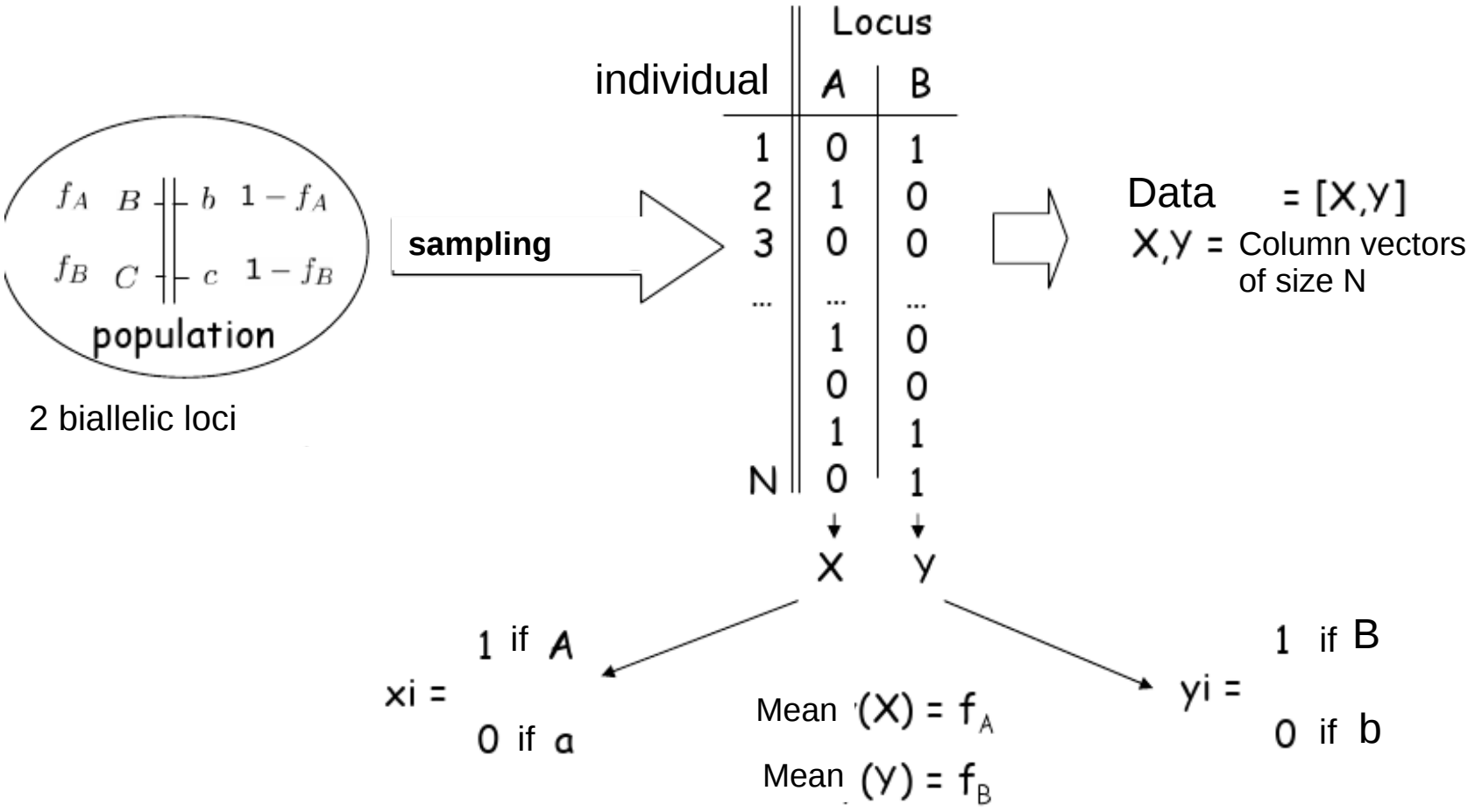
|D|=0; Dmax=0.22
|D'|=0; r^2=0

Site4 \ Site1	G	A
C	1	1
G	5	2

|D|=0.04; Dmax=0.15
|D'|=0.25; r^2=0.04

LD computations – matrix form

Matrix notation for LD



How to compute LD ? ... some linear algebra again...

Matrix notation for LD

individual	Locus A	Locus B
1	0	1
2	1	0
3	0	0
...	...	...
	1	0
	0	0
	1	1
N	0	1

Centered data $= [X^*, y^*]$

with $X^* = X - \text{Mean } X)$

$y^* = y - \text{Mean } y)$

$$Cov(X^*, y^*) = [X^*, y^*]T[X^*, y^*]$$

$$Cov(X^*, y^*) = \begin{pmatrix} f_A(1- f_A) & -D_{AB} \\ -D_{AB} & f_B(1- f_B) \end{pmatrix}$$

$$Cor(X^*, y^*) = \begin{pmatrix} 1 & -r^2_{AB} \\ -r^2_{AB} & 1 \end{pmatrix}$$ = LD matrix expressed as correlation between loci

LD within a locus

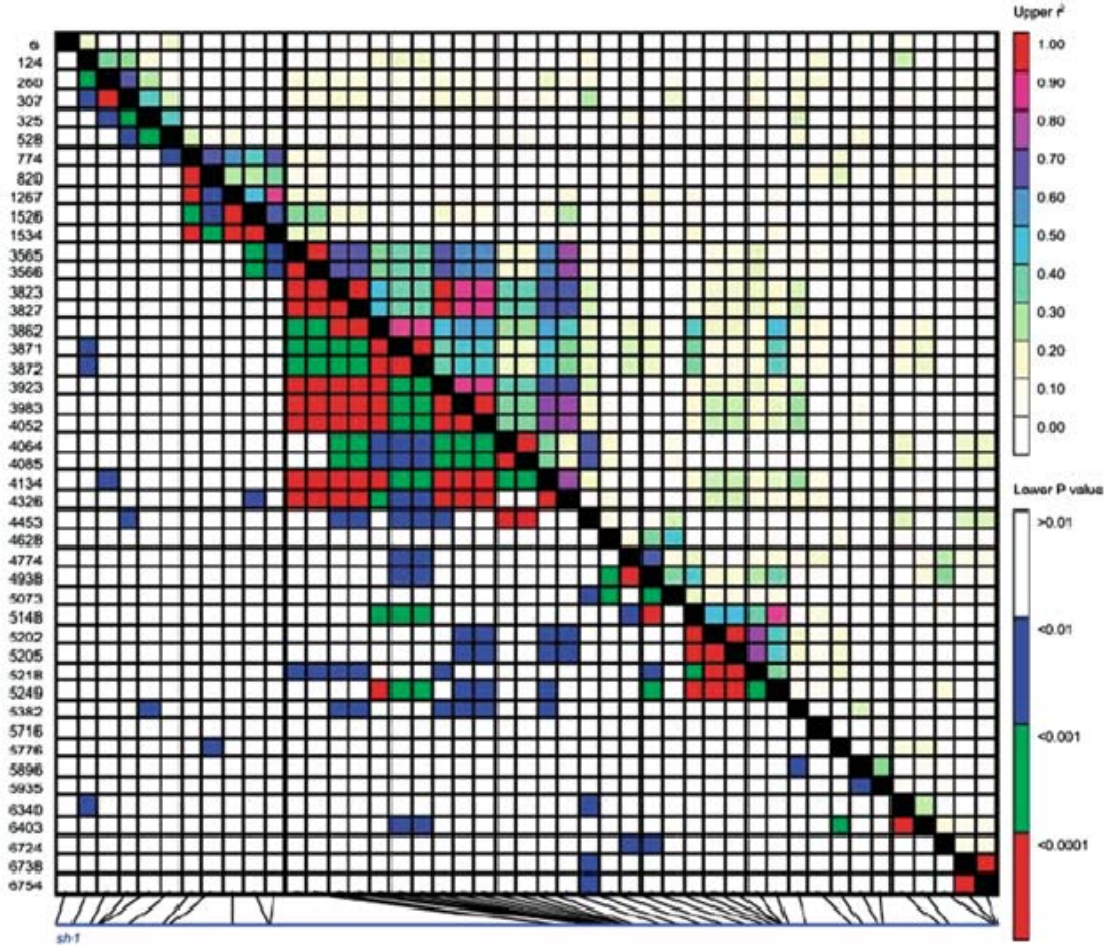
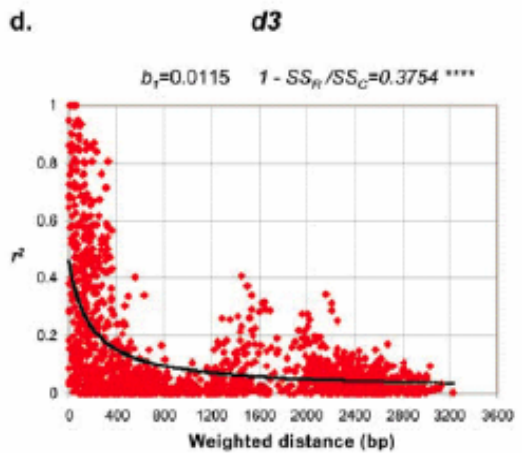
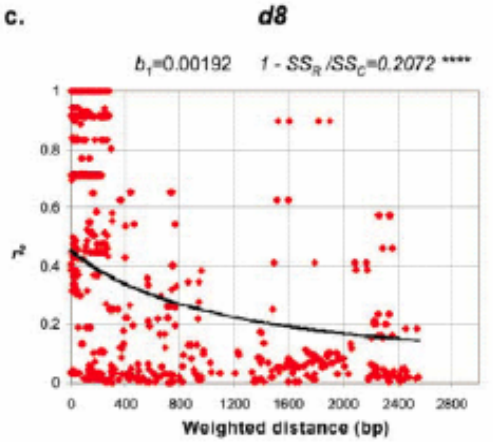


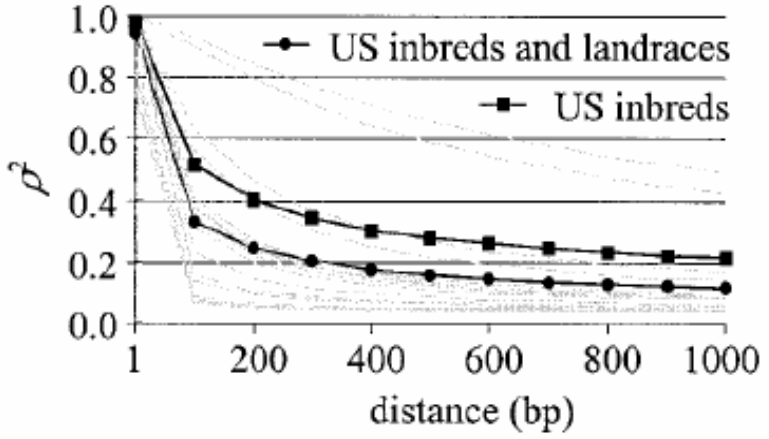
Figure 3. Disequilibrium matrix for polymorphic sites within *shrunken 1* (*sh1*). Polymorphic sites are plotted on both the X–axis and Y-axis. Pair-wise calculations of LD (r^2) are displayed above the diagonal with the corresponding *P*-values for Fisher's exact test displayed below the diagonal (reproduced with permission from Flint-Garcia *et al.*, 2003).

LD in corn





(Remington et al., 2001, PNAS 98:11479)



(Tenaillon et al., 2001, PNAS 98:9161)

LD diminue rapidement
varie entre genes et matériel génétique

What forces shape LD?

Natural and artificial selection

- Recombination rate
- Genetic drift
- Mutation rate
- Population structure
- Population expansion/bottleneck
- Admixture
- Mating system

How LD is created and evolve ?

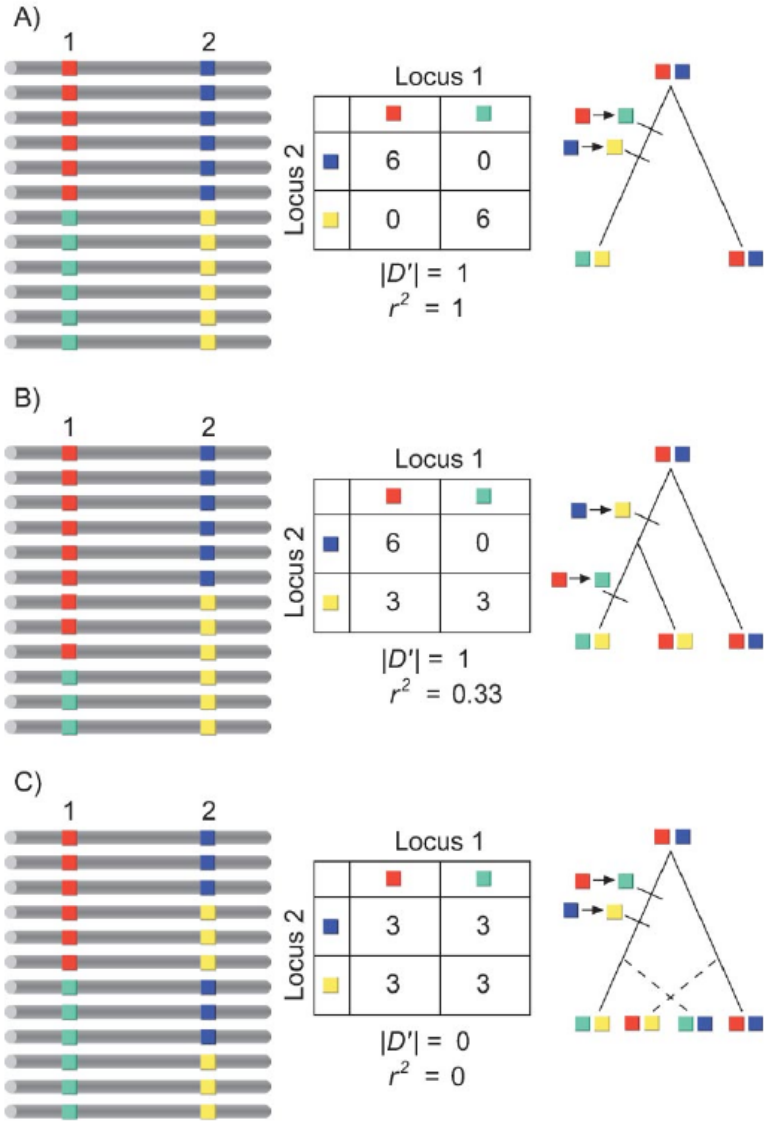


Figure 1 Hypothetical scenarios of linkage disequilibrium (LD) between linked polymorphisms caused by different mutational and recombinational histories demonstrating the behavior of the r^2 and D' statistics. Images in the *left* column represent the allelic states of two loci. The *middle* column represents the 2 × 2 contingency table of haplotypes and the resulting r^2 and D' statistics. The *right* column represents a possible tree responsible for the observed LD present. (*A*) Absolute LD exists when two loci share

Mutation and recombination : two opposed driving forces

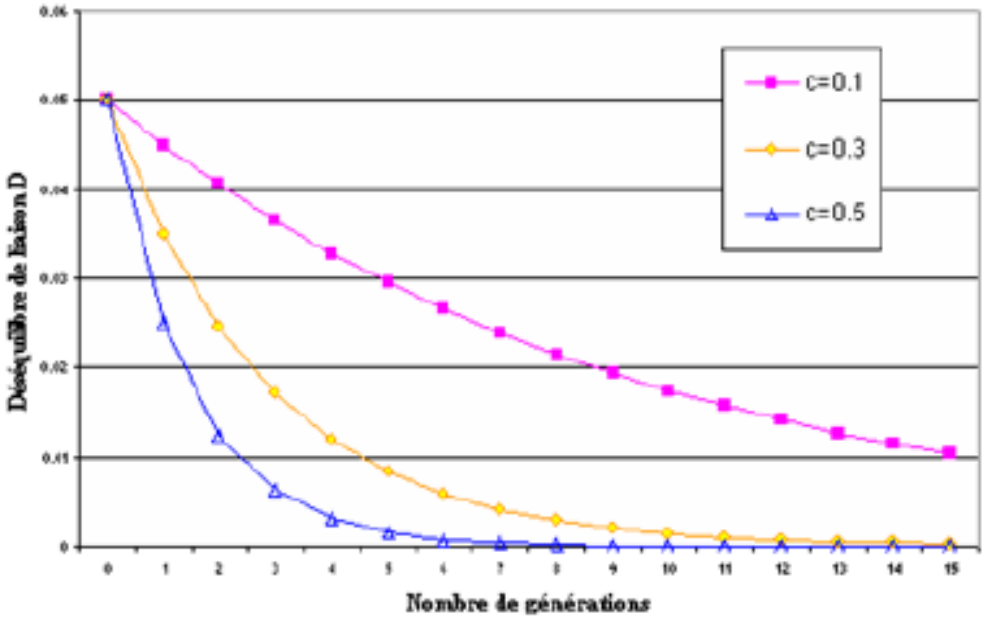


- Mutation : crée un LD — GA, AA, GG
- Recombinaison : détruit le LD — AG

LD decreases with time (generations)

$$D_{n+1}=D_n.(1-r)$$

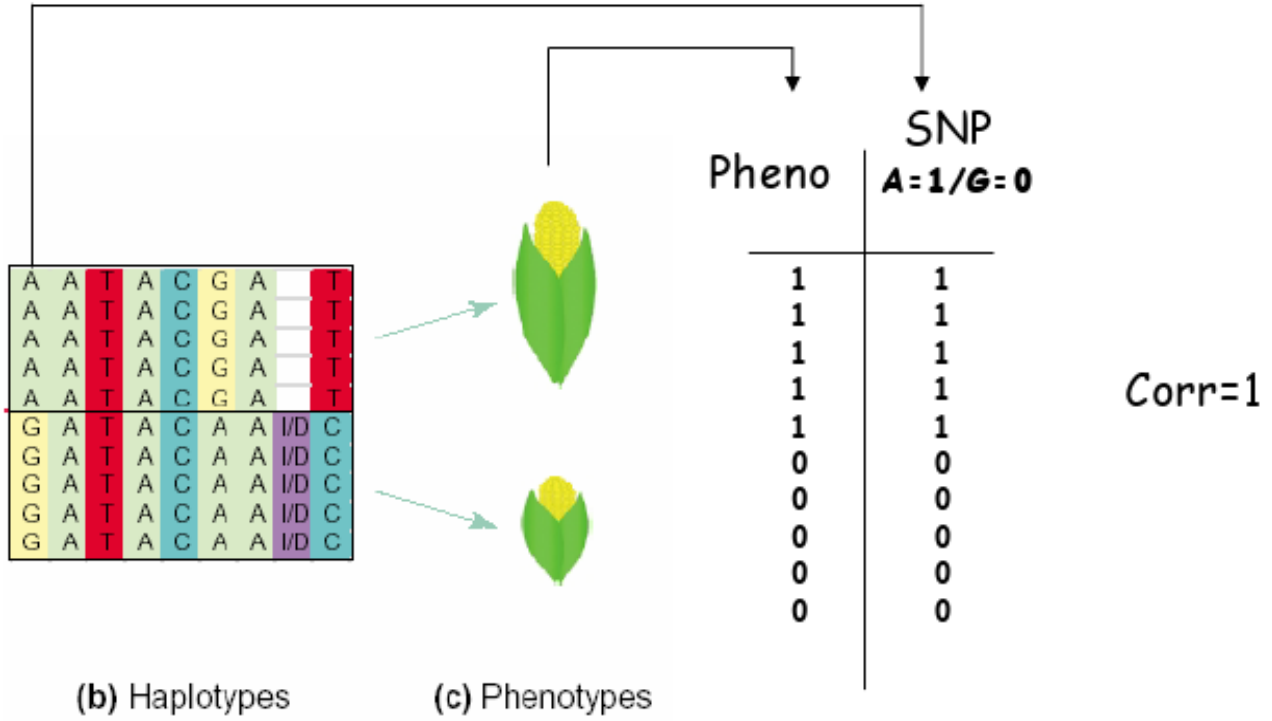
A. Sanchez-Mazas



In a panmictique population, recombination destroys LD

- difference between selfers and outcrossers

Trivial example of detecting an association



Looking for association : measurement of the correlation between phenotype and genotype

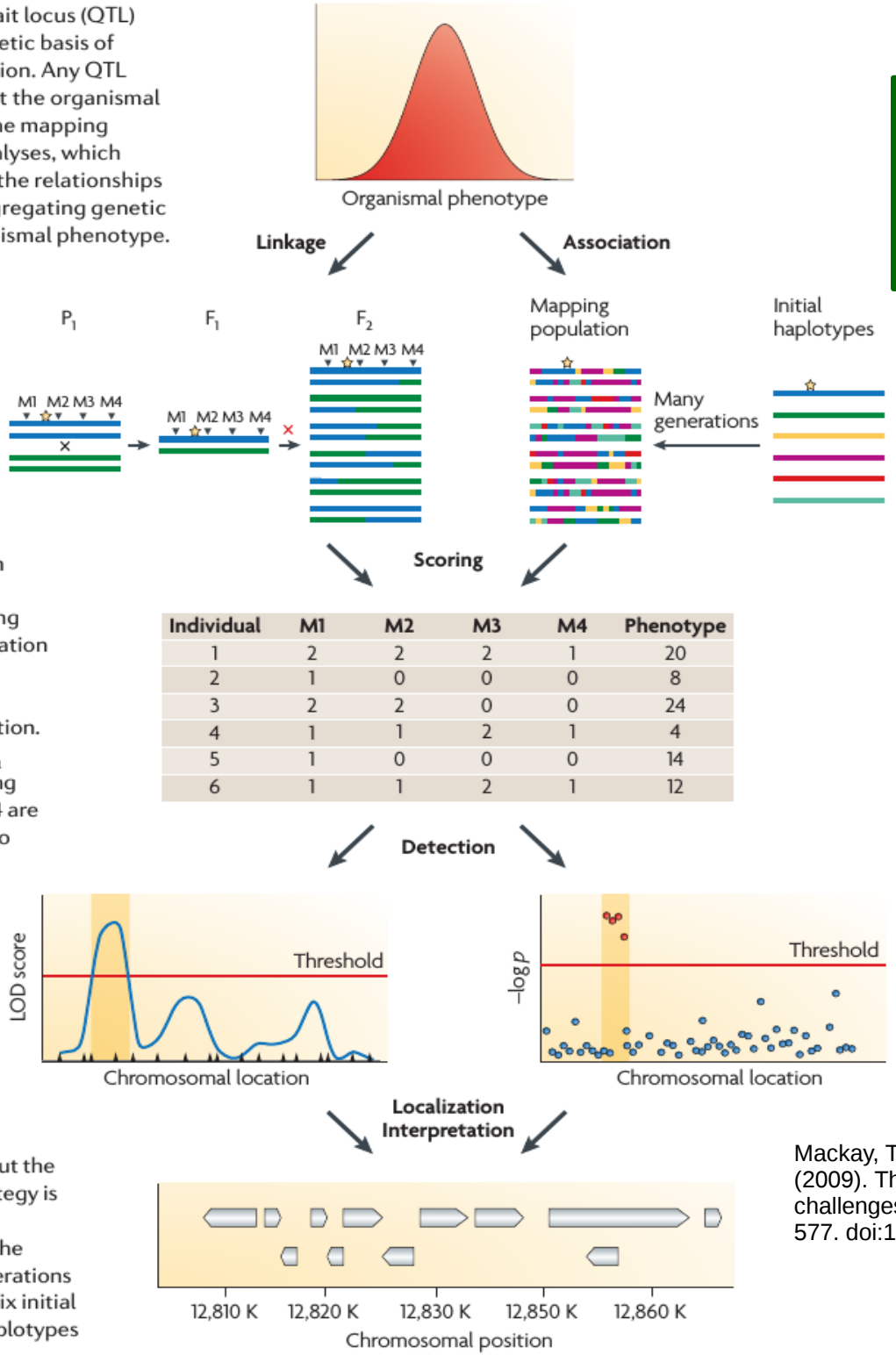
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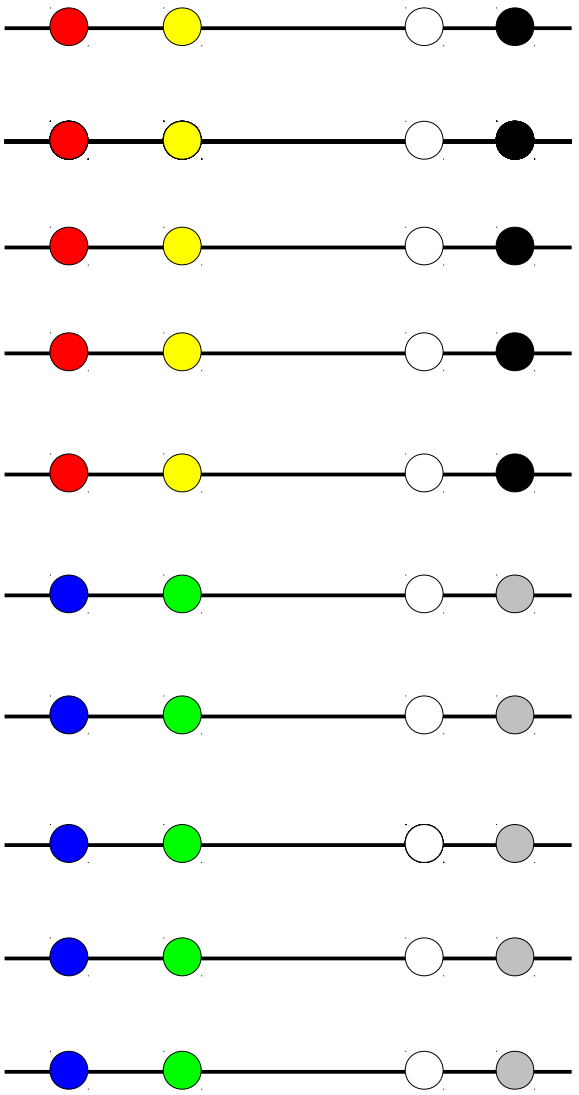


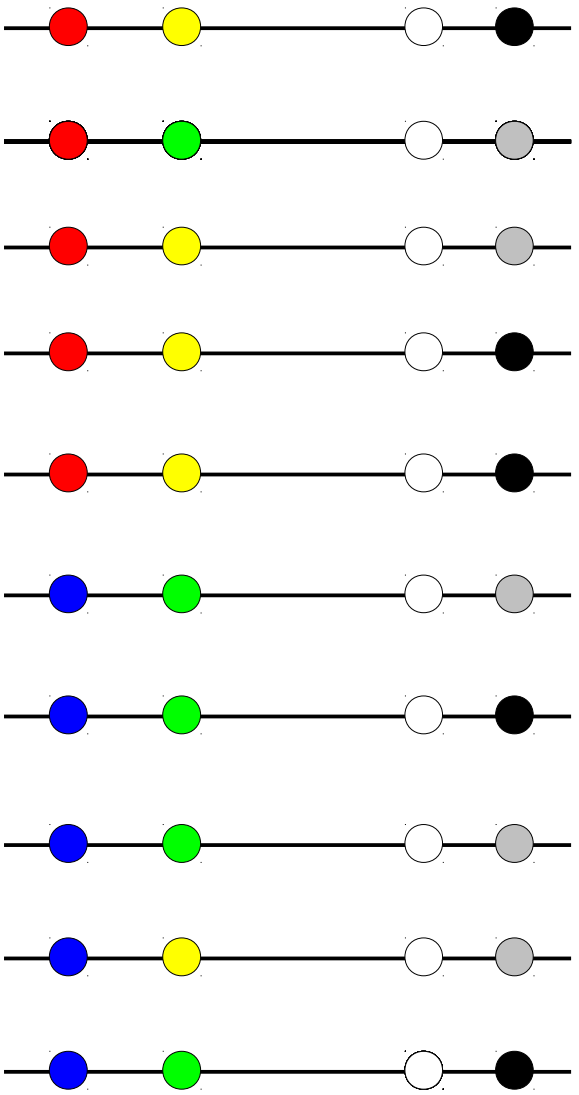
Relationship between ‘linkage genetics’ and ‘association genetics’

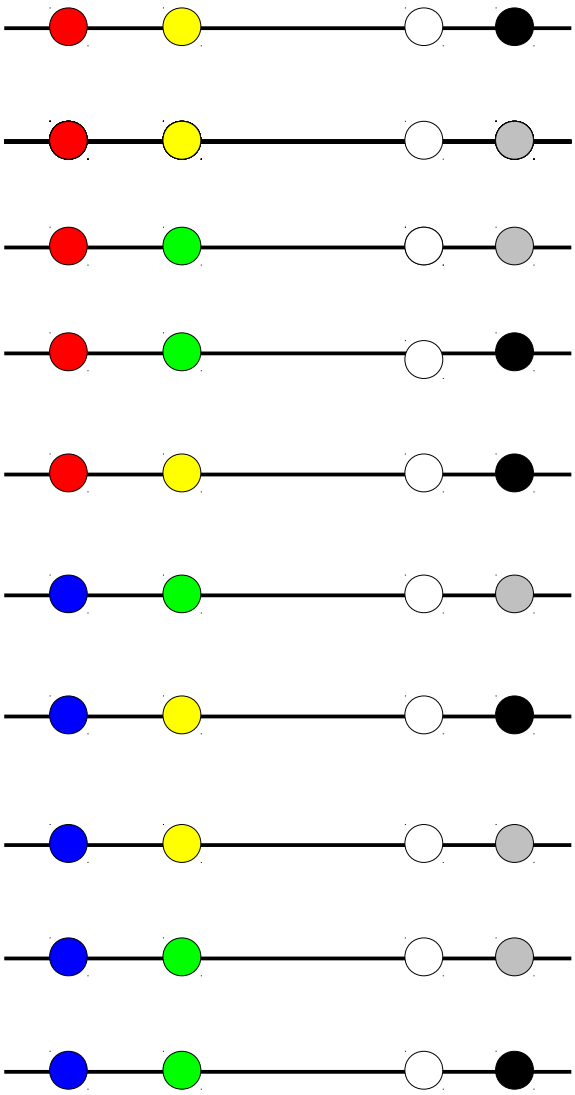
Mackay, T. F. C., Stone, E. A., and Ayroles, J. F. (2009). The genetics of quantitative traits: challenges and prospects. Nat Rev Genet 10, 565–577. doi:10.1038/nrg2612.

Linkage desiquilibrium (LD) : non random allele association along a chromosome









High LD

poor resolution

medium LD

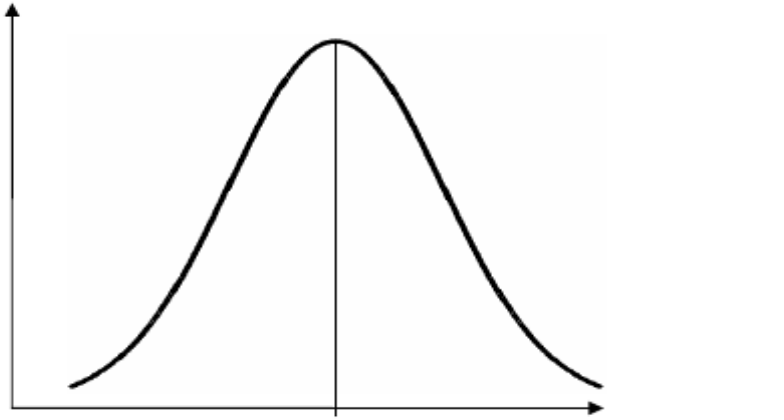
Accurate resolution

No LD

likely to miss

Does it works also for QTLs ?

Quantitative trait

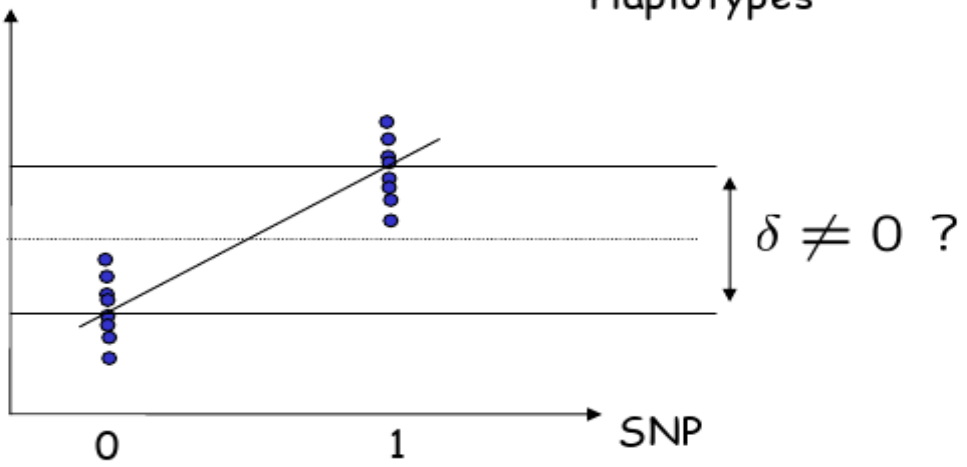


SNP1	SNP2	SNP3	SNP4	SNP5
1	1	1	0	1
0	1	0	1	0
0	0	1	0	0
0	1	0	1	1
1	0	0	1	0
0	1	0	0	1
0	0	1	1	0
1	1	1	1	1
1	1	1	0	1

Haplotypes

ANOVA to compare means

$Yi : \text{mean} \cdot a^{*}SNP + ei$



r^2 in association mapping

Explains 10% of total phenotypic variance
NOT Genotyped Causative SNP



Locus 2 \ Locus 1	●	●
○	5	3
○	0	2

$|D'| = 1$

$r^2 = 0.25$

SNP Marker will explain 25% of the total QTL variation, but only 2.5% of the total phenotypic variation. Need large sample size.

r^2> 0.80 is recommended for association studies

Practical using TASSEL 5.2

- TASSEL
  - Running the software
  - Loading and exploring SNPs data

- Computations of LD plots

At your keyboards

What are the main drawbacks of association genetics ?

- Lack of chance in sampling
- Lack of knowledge of population structure
- Admixture

Sub-populations : a key problem that increase false-positives

- 400 Scandinavians and Spaniards combined

	B_1	B_2
A_1	320	200
A_2	200	80

$\chi^2 = 7.81$

Pop. Structure ignored

Pop. Structure accounted

- 200 Scandinavians

	B_1	B_2
A_1	160	160
A_2	40	40

$\chi^2 = 0$

- 200 Spaniards

	B_1	B_2
A_1	160	40
A_2	160	40

$\chi^2 = 0$

Sub-populations : a key problem that increase false-positives

- 400 Scandinavians and Spaniards combined

	B_1	B_2
A_1	320	200
A_2	200	80

$\chi^2 = 7.81$

- Spurious association
  - not reflective of genetic distance
    - *A* and *B* might be on different chromosomes

Pop. Structure ignored

Pop. Structure accounted

- 200 Scandinavians

	B_1	B_2
A_1	160	160
A_2	40	40

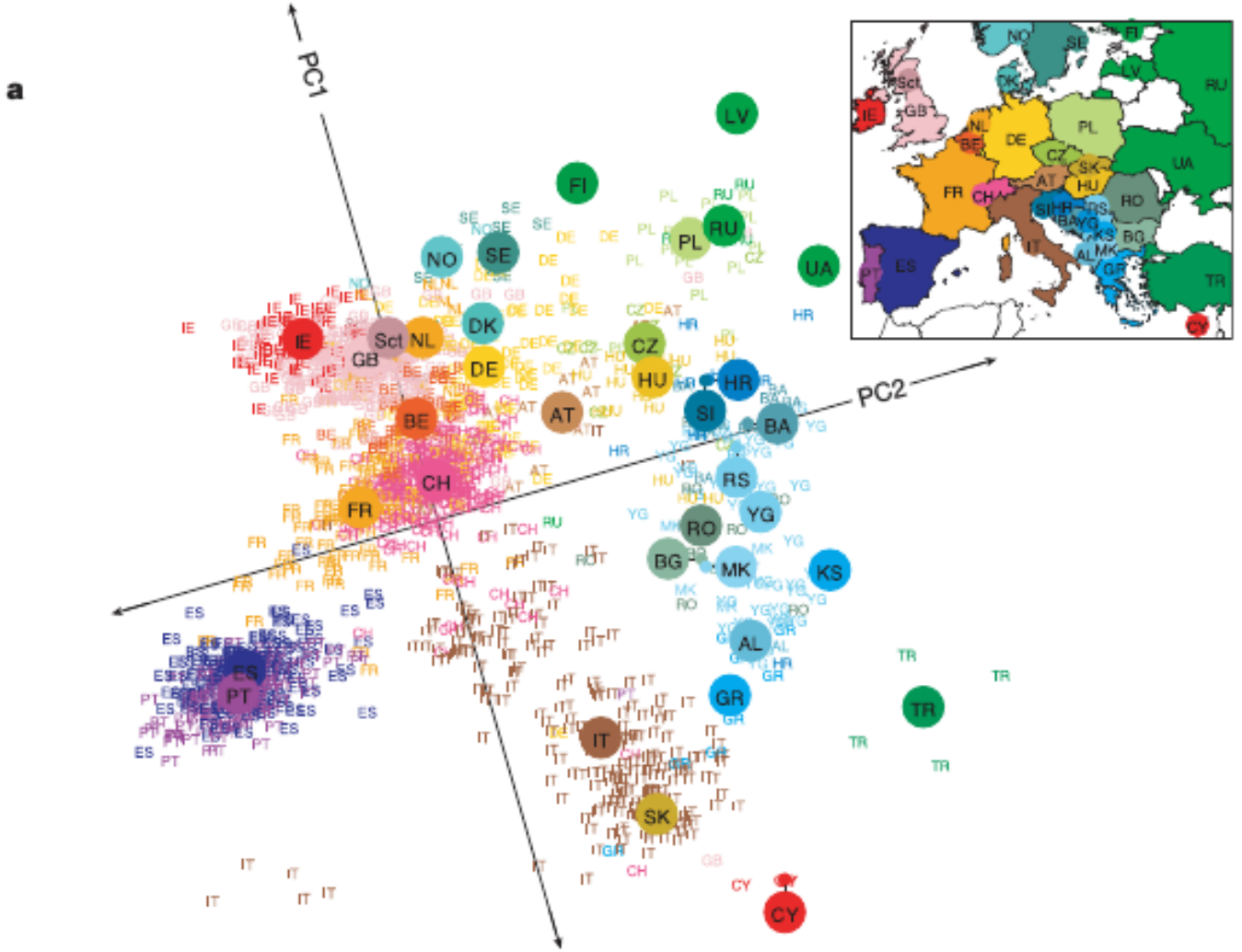
$\chi^2 = 0$

- 200 Spaniards

	B_1	B_2
A_1	160	40
A_2	160	40

$\chi^2 = 0$

Genes, geography and populations



Novembre, J., Johnson, T., Bryc, K., Kutalik, Z., Boyko, A. R., Auton, A., Indap, A., King, K. S., Bergmann, S., Nelson, M. R., et al. (2008). Genes mirror geography within Europe. Nature 456, 98–101. doi:10.1038/nature07331.

Population Structure in Crops

TABLE 1

Pairwise F_{ST} values and AMOVA

	Aromatic	*Aus*	*Indica*	*Temperate japonica*	*Tropical japonica*
Aromatic		0.34	0.39	0.37	0.31
Aus	0.28		0.26	0.44	0.37
Indica	0.34	0.25		0.45	0.39
Temperate japonica	0.29	0.42	0.43		0.22
Tropical japonica	0.23	0.35	0.36	0.20	

AMOVA-based estimates appear above the diagonal and classical estimates (θ) appear below the diagonal. For AMOVA-based estimates, $P < 0.000001$ for 110 permutations for all population comparisons.

Garris et al. (2005) Genetics 169: 1631–1638.

Table 2 Fst values for comparison of subpopulations determined by STRUCTURE analysis

Fst*	Mixed	Non-stiff stalk	Popcorn	Stiff stalk	Sweetcorn	Tropical/subtropical
Mixed		0.01	0.15	0.15	0.08	0.03
Non-stiff stalk			0.15	0.18	0.10	0.06
Popcorn				0.39	0.28	0.16
Stiff stalk					0.33	0.22
Sweetcorn						0.12
Tropical/subtropical						

*All Fst values are significant ($P < 0.001$).

Flint-Garcia et al. (2005) Plant Journal 44: 1054–1064.

Correlation between Population Structure and Traits

Table 1 Descriptive statistics, family mean-basis heritability and percentage of phenotypic variation explained by population structure for 60 traits scored at up to 10 environments

	Minimum	Average	Maximum	Percentage of range captured by B73 and Mo17	Significance of G × E	H_m^{2a}	h_p^{2b}	R^{2c}
Plant traits								
Days to pollen (days)	50.2	70.1	90.8	17.9	**	0.965	0.877	35.0
Days to silk (days)	51.6	70.7	90.2	14.9	**	0.956	0.847	32.9
Plant height (cm)	86	154	211	8.8	**	0.949	0.739	8.7
Ear height (cm)	20	57	100	15.7	**	0.952	0.800	16.4
Upper leaf angle (°)	6.0	54.4	80.6	32.2	ND	0.881	0.726	20.1
Lower leaf angle (°)	31.9	55.6	76.2	16.7	ND	0.860	0.533	6.7
Leaf length (mm)	44.6	74.6	104.2	30.2	ND	0.936	0.920	17.6
Leaf width (cm)	4.4	8.5	11.6	2.6	ND	0.891	0.710	4.6
Nodes with brace roots	0.1	1.7	4.3	27.7	ND	0.773	0.593	12.2
Number of ears[d]	1.2	2.0	3.2	6.4	ND	0.639	0.195	5.0
Nodes from ear to tassel	2.9	5.4	8.4	20.2	ND	0.930	0.859	14.5
Nodes below ear	2.8	5.9	9.8	27.1	ND	0.913	0.817	22.1
Total number of nodes	7.1	13.3	18.8	24.7	ND	0.948	0.911	24.5
Stalk thickness (mm)	15.4	21.6	27.5	25.2	ND	0.811	0.541	1.1
Stalk width (mm)	15.7	21.8	28.2	25.1	ND	0.786	0.532	1.7
Tassel length (mm)	188.3	321.0	474.9	22.8	ND	0.900	0.763	17.5
Main spike length (mm)	119.0	220.5	370.4	35.4	ND	0.912	0.710	3.7

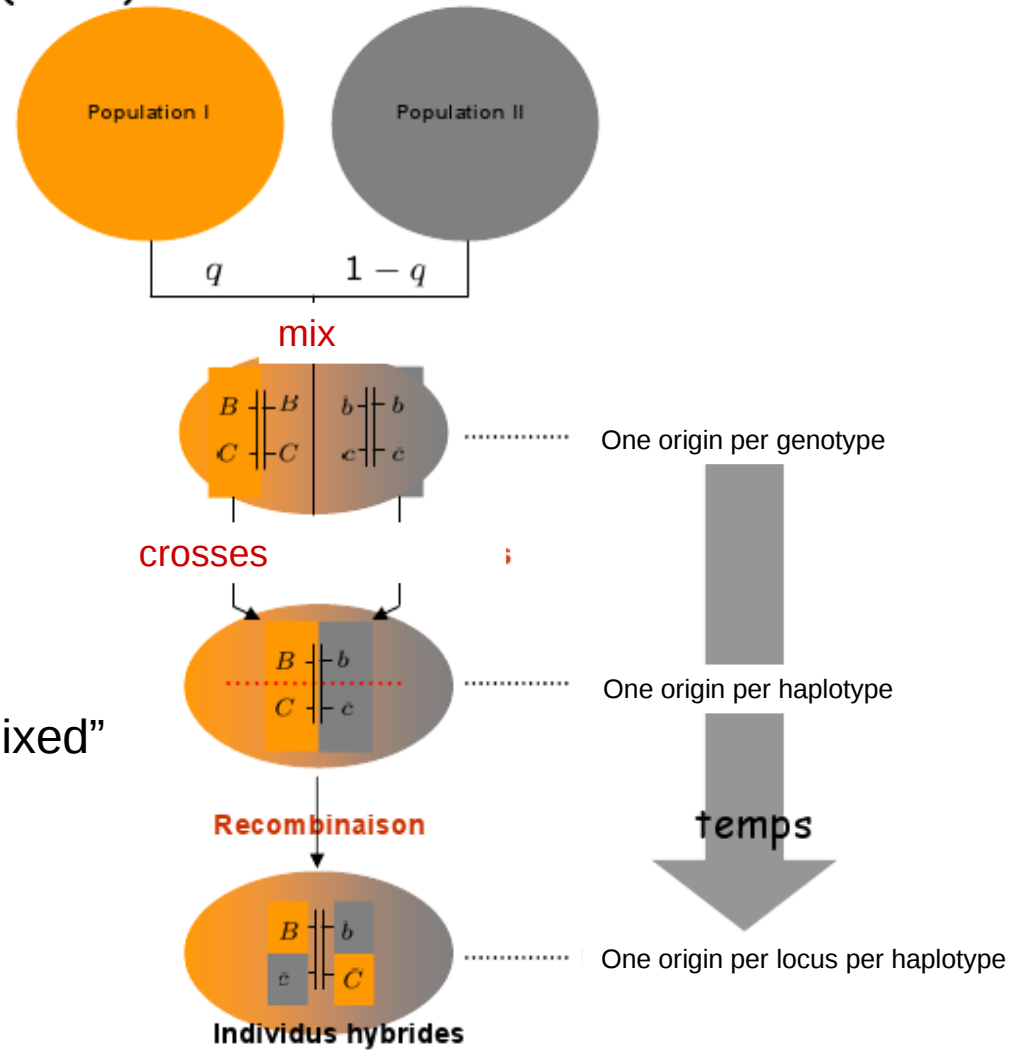
Flint-Garcia et al. (2005) Plant Journal 44: 1054–1064.

Traits may be the cause of population structure

→ If population structure has a strong correlation with phenotype, knowing the origin of a sample (its admixture components) may help to infer its phenotype

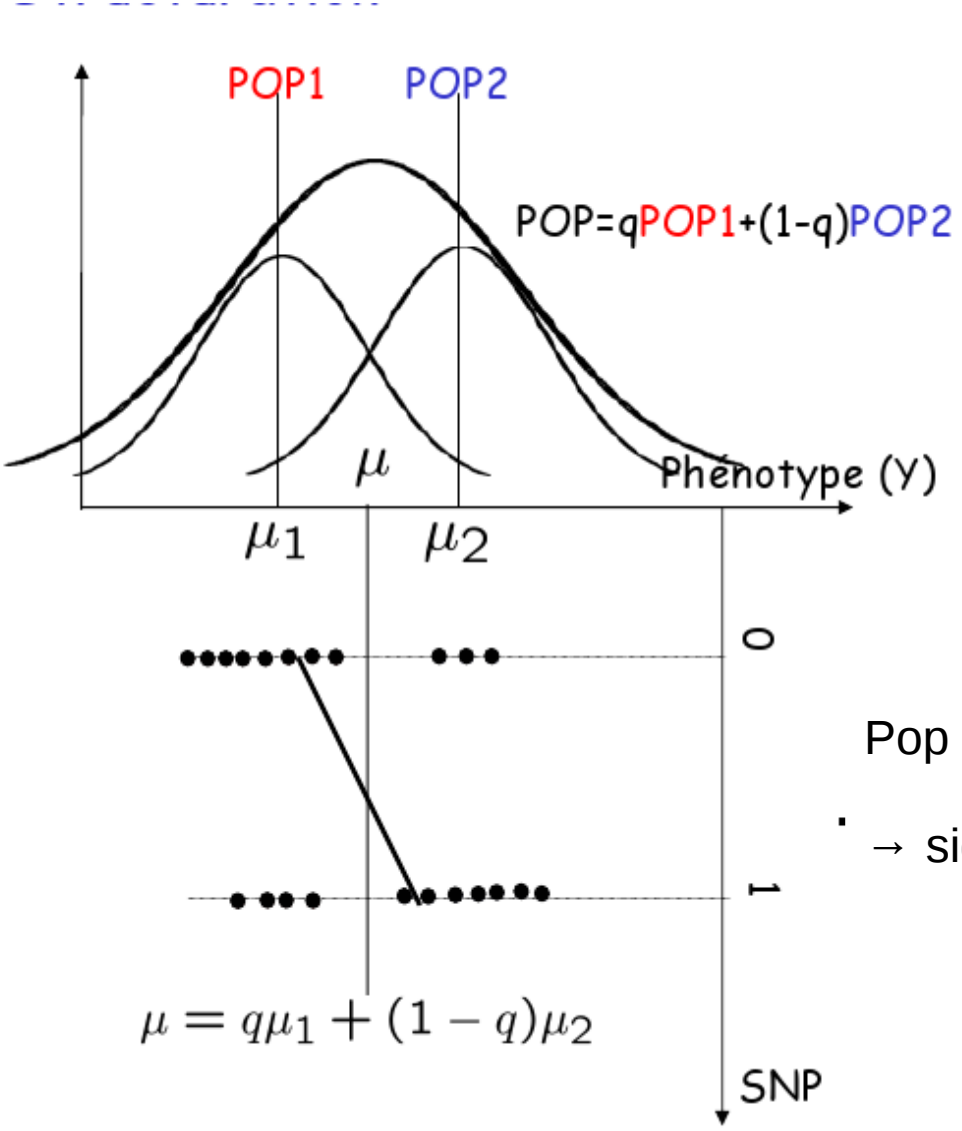
Admixed populations : another problem

Hybrid Isolation Model (HIM)



Hybrid population or “admixed” = a mix of genome from various origins

Pop. structure and quantitative trait



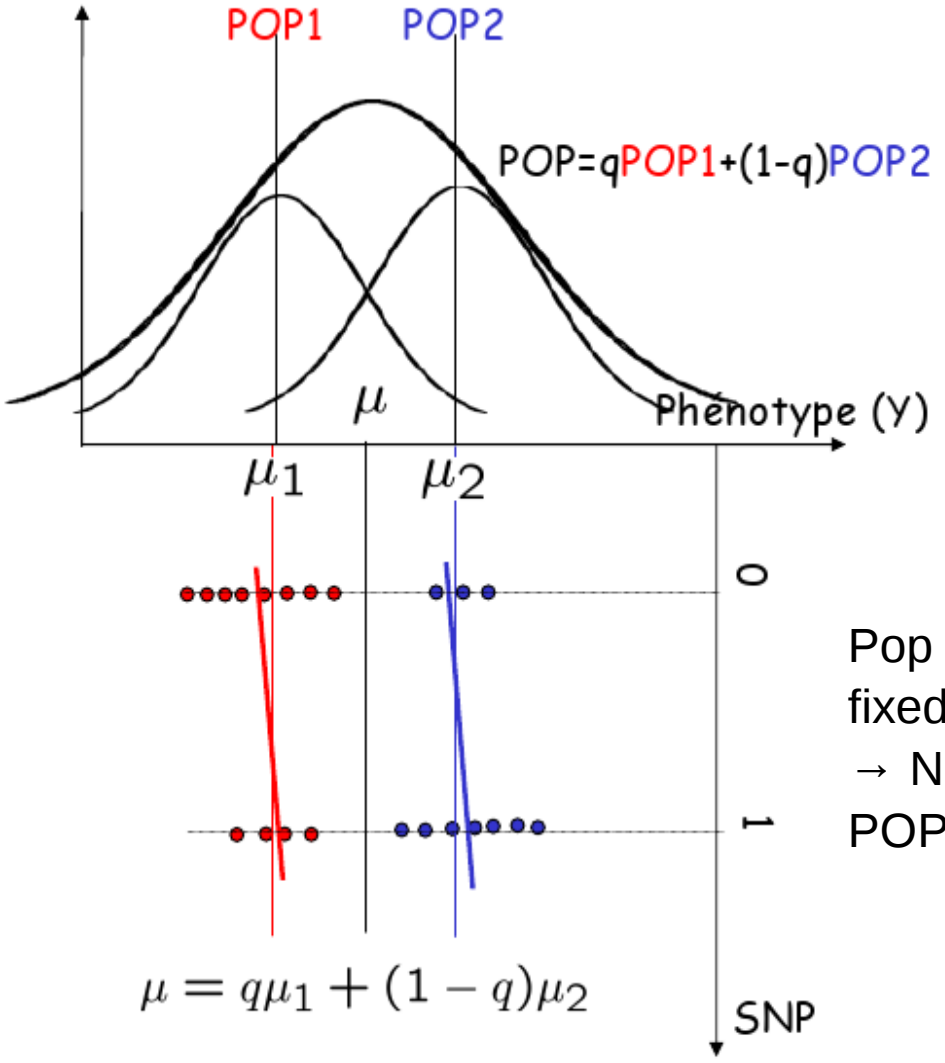
SNP1	SNP2	SNP3	SNP4
1	1	1	0
0	1	0	1
0	0	1	0
0	1	0	1
1	0	0	1
0	1	0	0
0	0	1	1
1	1	1	1
1	1	1	0

Haplotypes + Population

Pop structure ingored/unknown :

→ significant effet of SNP in POP

Pop. structure and quantitative trait



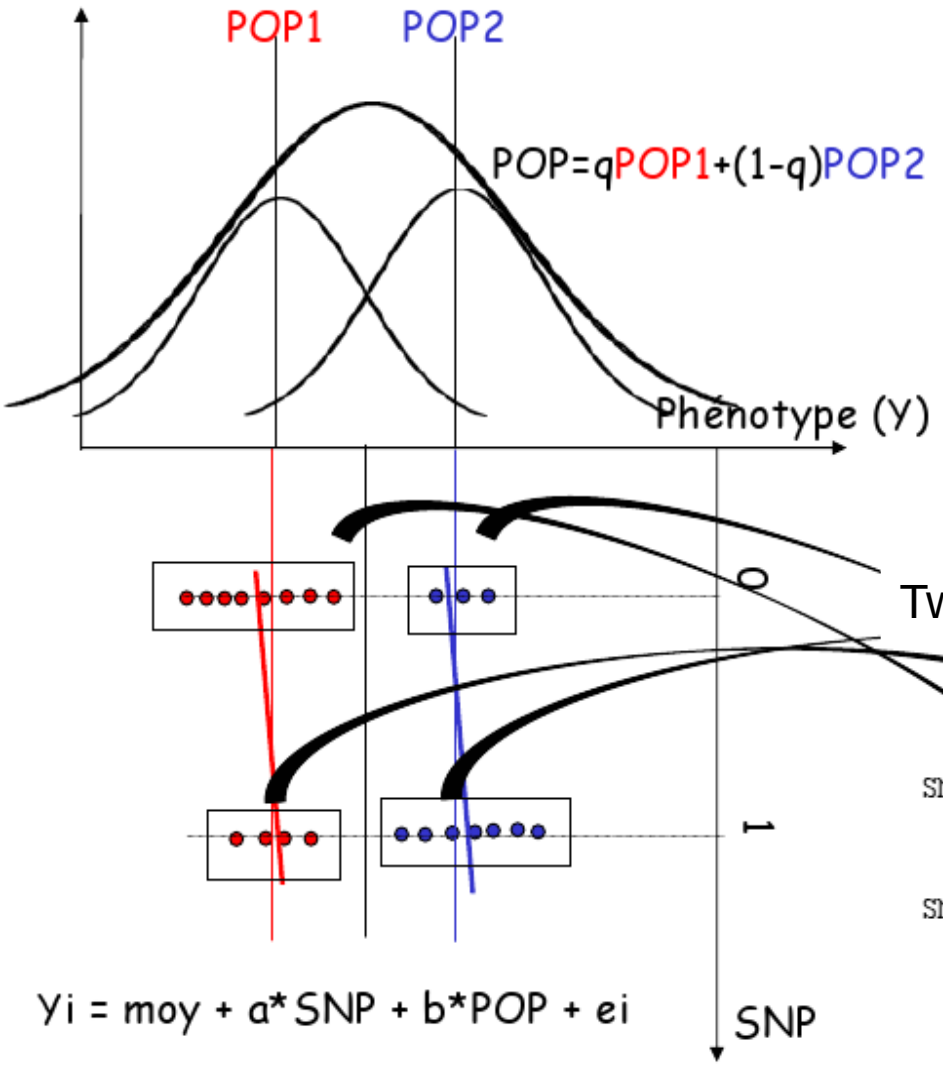
SNP1	SNP2	SNP3	SNP4	POP
1	1	1	0	1
0	1	0	1	1
0	0	1	0	2
0	1	0	1	2
1	0	0	1	2
0	1	0	0	1
0	0	1	1	2
1	1	1	1	1
1	1	1	0	1

Haplotypes + Population

Pop structure <u>included in the model</u> as fixed effect

→ No effet of SNP within each POP1/POP2

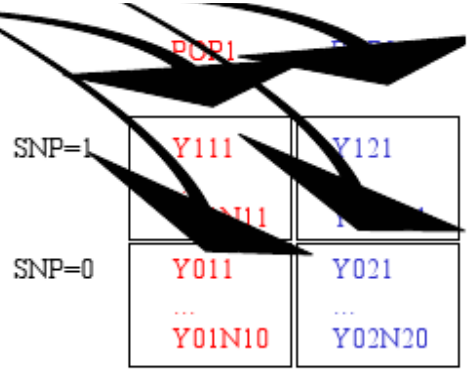
Pop. structure and quantitative traits



SNP1	SNP2	SNP3	SNP4	POP
1	1	1	0	1
0	1	0	1	1
0	0	1	0	2
0	1	0	1	2
1	0	0	1	2
0	1	0	0	1
0	0	1	1	2
1	1	1	1	1
1	1	1	0	1

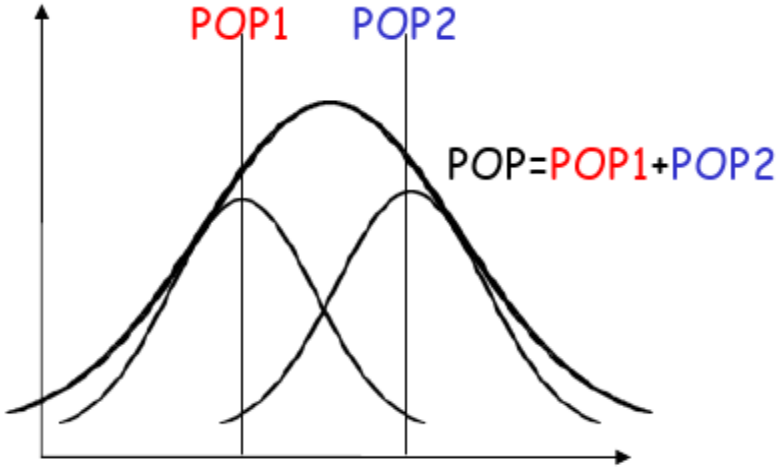
Haplotypes + Population

Two-way ANOVA : SNP + POP



Pop. structure and quantitative traits

Structure probabiliste



	SNP1	SNP2	SNP3	SNP4	POP1	POP2
	1	1	1	0	0.2	0.8
	0	1	0	1	0.5	0.5
	0	0	1	0	0.3	0.7
i	0	1	0	1	0.2	0.8
	1	0	0	1	0.6	0.4
	0	1	0	0	0.9	0.1
	0	0	1	1	1.0	0.0
	1	1	1	1	0.7	0.3
	1	1	1	0	0.4	0.6

Haplotypes + Population

Linear model :

$$Yi = moy + a^* SNP + b^*(Prob(i->POP1) - Prob(i->POP2)) + ei$$

Prob(i->POPj) : Probability that sample i comes from POPj (j = 1,2)

Linear Mixed Model

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Zu} + \mathbf{e}$$

$$\mathbf{y} = X\boldsymbol{\beta} + S\boldsymbol{\alpha} + Q\mathbf{v} + \mathbf{Zu} + \mathbf{e}$$

y is a vector of phenotypic observation
β is a vector of fixed effects other than SNP or population group effects;
α is a vector of SNP effects (QTN);
v is a vector of population effects;
u is a vector of polygene background effects;
e is a vector of residual effects;
Q is a matrix from STRUCTURE relating **y** to **v**; and
X, **S** and **Z** are incidence matrices of 1s and 0s relating **y** to **β**, **α** and **u**, respectively.

Yu, Pressoir, et al. 2005. Nature Genetics 38:203-208

TASSEL implements the LMMs with several tricks (CLMM, P3D, ….) to reduce computation time
User manual of TASSEL is worth to read

More details for LMM

See other document ...

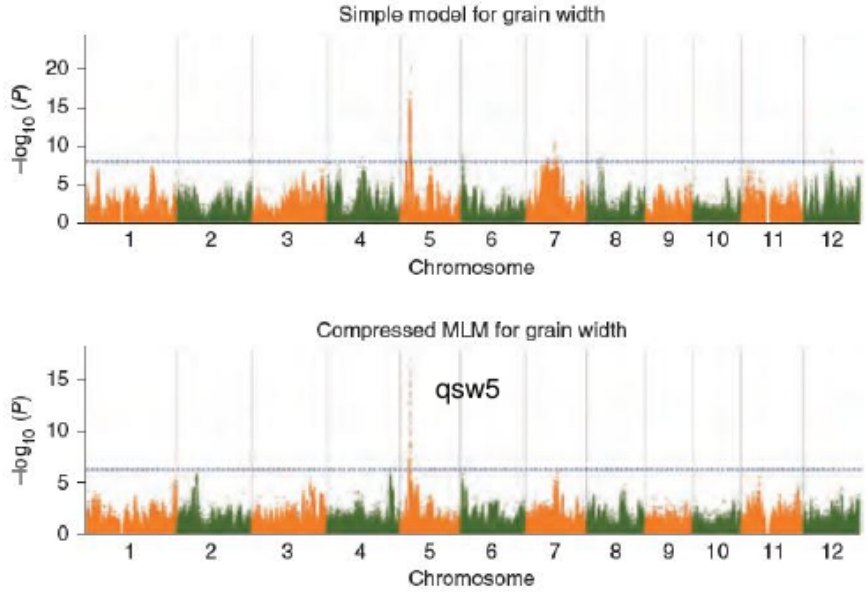
Practical with TASSEL 5.2

- TASSEL - following

- GWAS for Aphanomyces

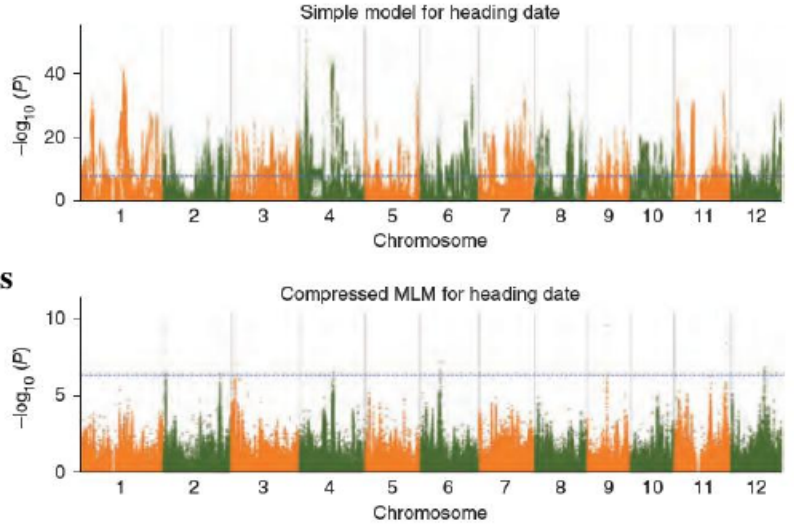
At your keyboards !

GWAS in 373 indica rice lines with nearly 1 million SNPs: low structure



Huang et al. 2010. Nature Genetics 42:961-967

GWAS in 373 indica rice lines with nearly 1 million SNPs: high structure



Need crosses

Huang et al. 2010. Nature Genetics 42:961-967

An *Arabidopsis* Example of Association Mapping in Structured Samples

Keyan Zhao[1], María José Aranzana[1], Sung Kim[1], Clare Lister[2], Chikako Shindo[2], Chunlao Tang[1], Christopher Toomajian[1], Honggang Zheng[1], Caroline Dean[2], Paul Marjoram[3], Magnus Nordborg[1*]

1 Molecular and Computational Biology, University of Southern California, Los Angeles, California, United States of America, **2** Department of Cell and Developmental Biology, John Innes Centre, Norwich, United Kingdom, **3** Department of Preventive Medicine, Keck School of Medicine, University of Southern California, Los Angeles, California, United States of America

A potentially serious disadvantage of association mapping is the fact that marker-trait associations may arise from confounding population structure as well as from linkage to causative polymorphisms. Using genome-wide marker data, we have previously demonstrated that the problem can be severe in a global sample of 95 *Arabidopsis thaliana* accessions, and that established methods for controlling for population structure are generally insufficient. Here, we use the same sample together with a number of flowering-related phenotypes and data-perturbation simulations to evaluate a wider range of methods for controlling for population structure. We find that, in terms of reducing the false-positive rate while maintaining statistical power, a recently introduced mixed-model approach that takes genome-wide differences in relatedness into account via estimated pairwise kinship coefficients generally performs best. By combining the association results with results from linkage mapping in F2 crosses, we identify one previously known true positive and several promising new associations, but also demonstrate the existence of both false positives and false negatives. Our results illustrate the potential of genome-wide association scans as a tool for dissecting the genetics of natural variation, while at the same time highlighting the pitfalls. The importance of study design is clear; our study is severely under-powered both in terms of sample size and marker density. Our results also provide a striking demonstration of confounding by population structure. While statistical methods can be used to ameliorate this problem, they cannot always be effective and are certainly not a substitute for independent evidence, such as that obtained via crosses or transgenic experiments. Ultimately, association mapping is a powerful tool for identifying a list of candidates that is short enough to permit further genetic study.

Citation: Zhao K, Aranzana MJ, Kim S, Lister C, Shindo C, et al. (2007) An *Arabidopsis* example of association mapping in structured samples. PLoS Genet 3(1): e4. doi:10.1371/journal.pgen.0030004

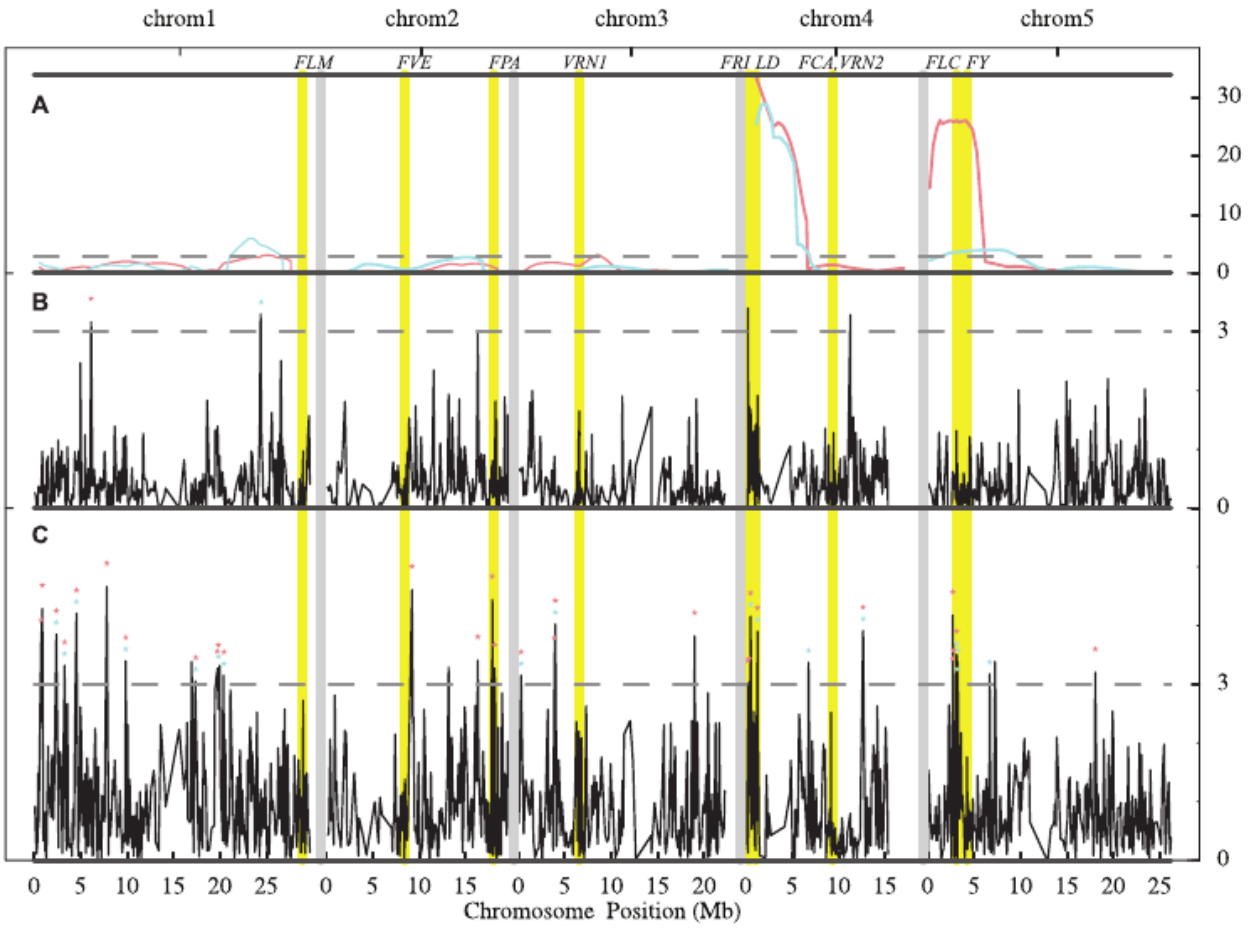
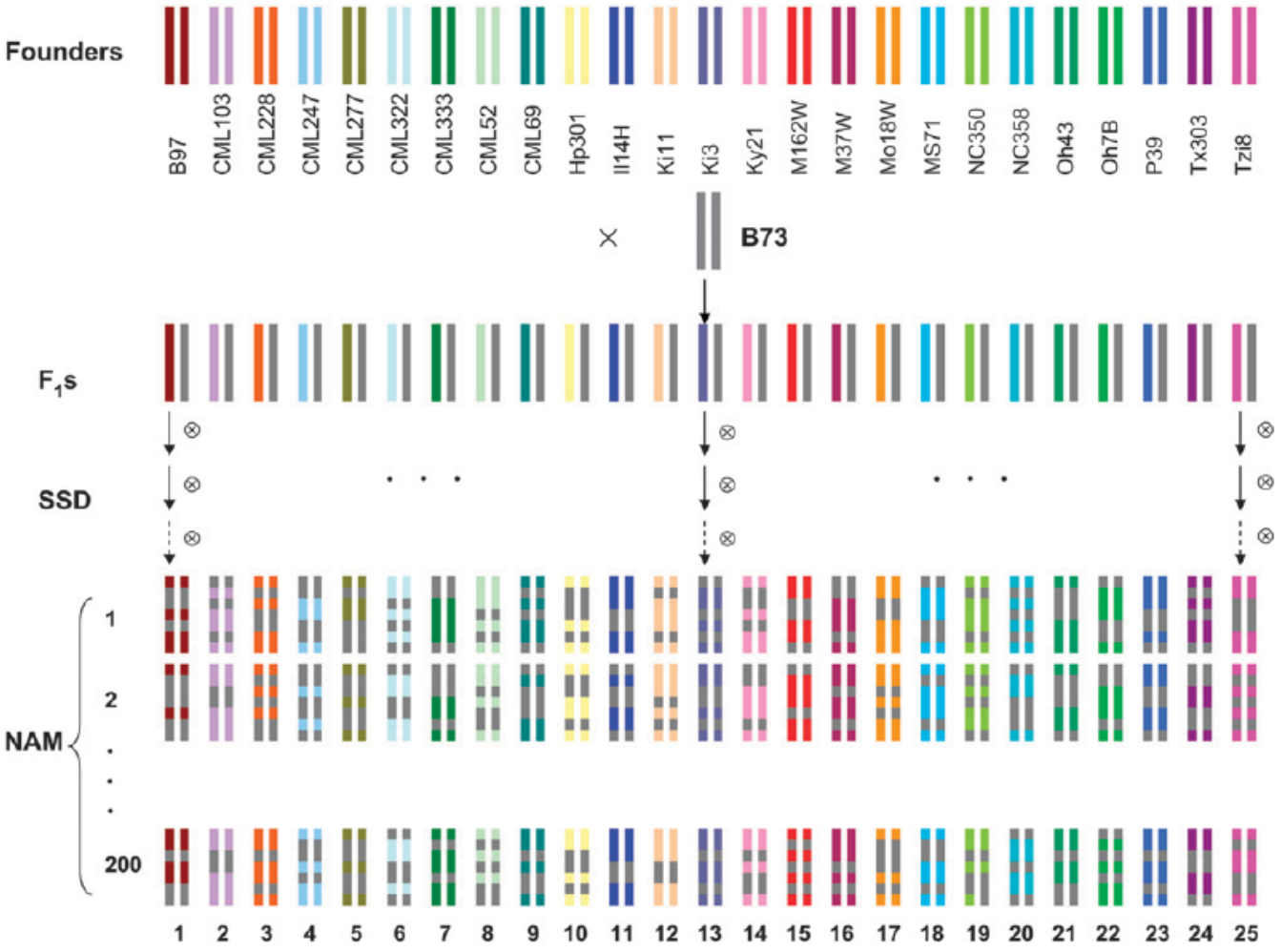


Figure 5. Comparison of Association and Linkage Mapping Results for the Phenotype JIC4W

The position of candidate genes with higher marker density are highlighted in yellow.
(A) The QTL LOD scores for two crosses: *Col-0* × *Lov-1* (pink curve) and *Col-0* × *Edi-0* (cyan curve). The dashed grey line marks a genome-wide permutation significance level of 5%.
(B) Negative log *p*-values from a genome-wide scan using the **Q** + **K** model.
(C) Negative log *p*-values from a genome-wide scan using a naive Kruskal-Wallis approach.
(B and C) The dashed grey line marks a nominal significance level of 0.1%, and the colored stars denote associations that should correspond to QTL in the cross displayed using the same color (because the appropriate alleles are segregating; see text).
doi:10.1371/journal.pgen.0030004.g005

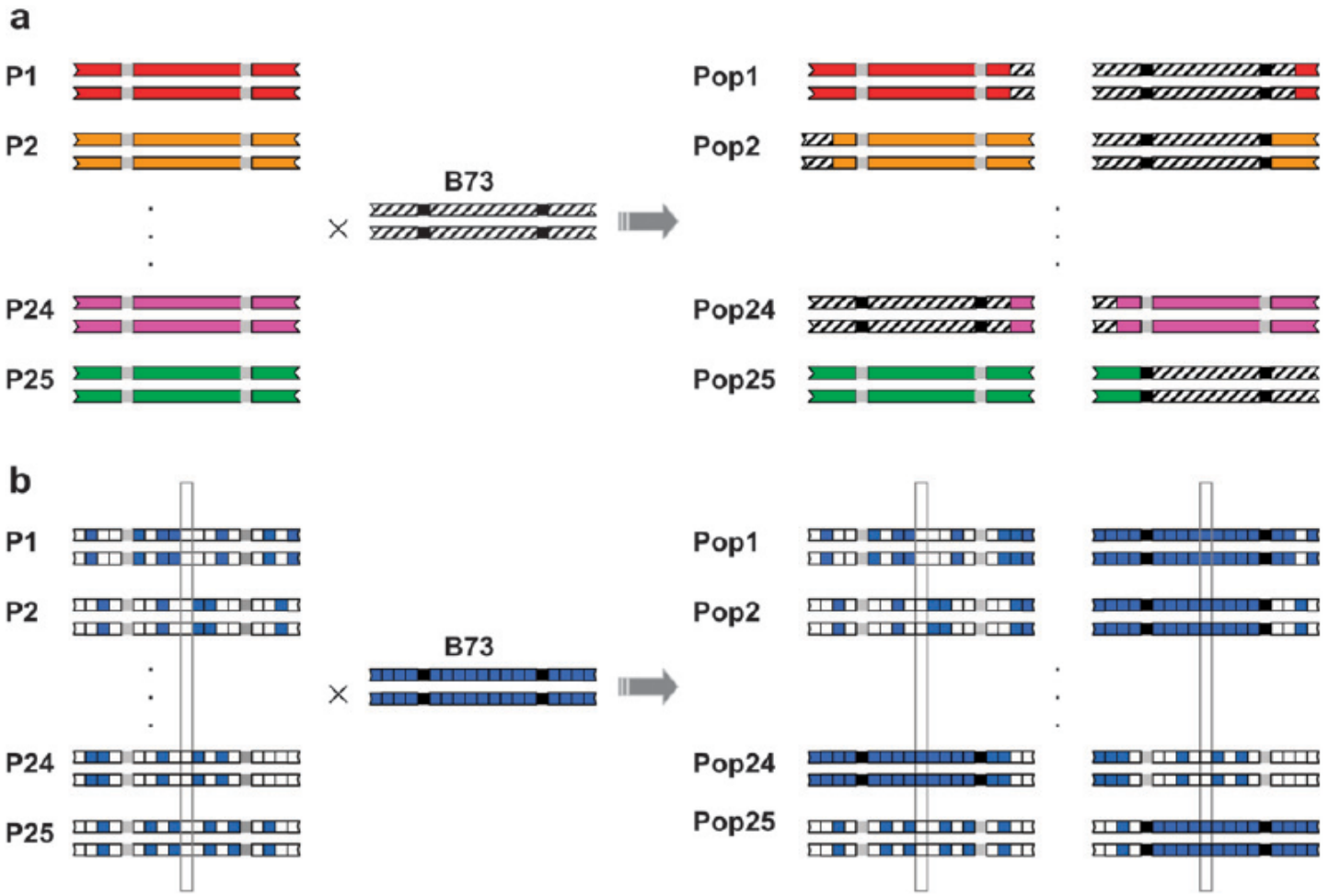
Integration of Linkage Analysis and GWAS for Trait Dissection



25 families for a total of 5,000 RILs

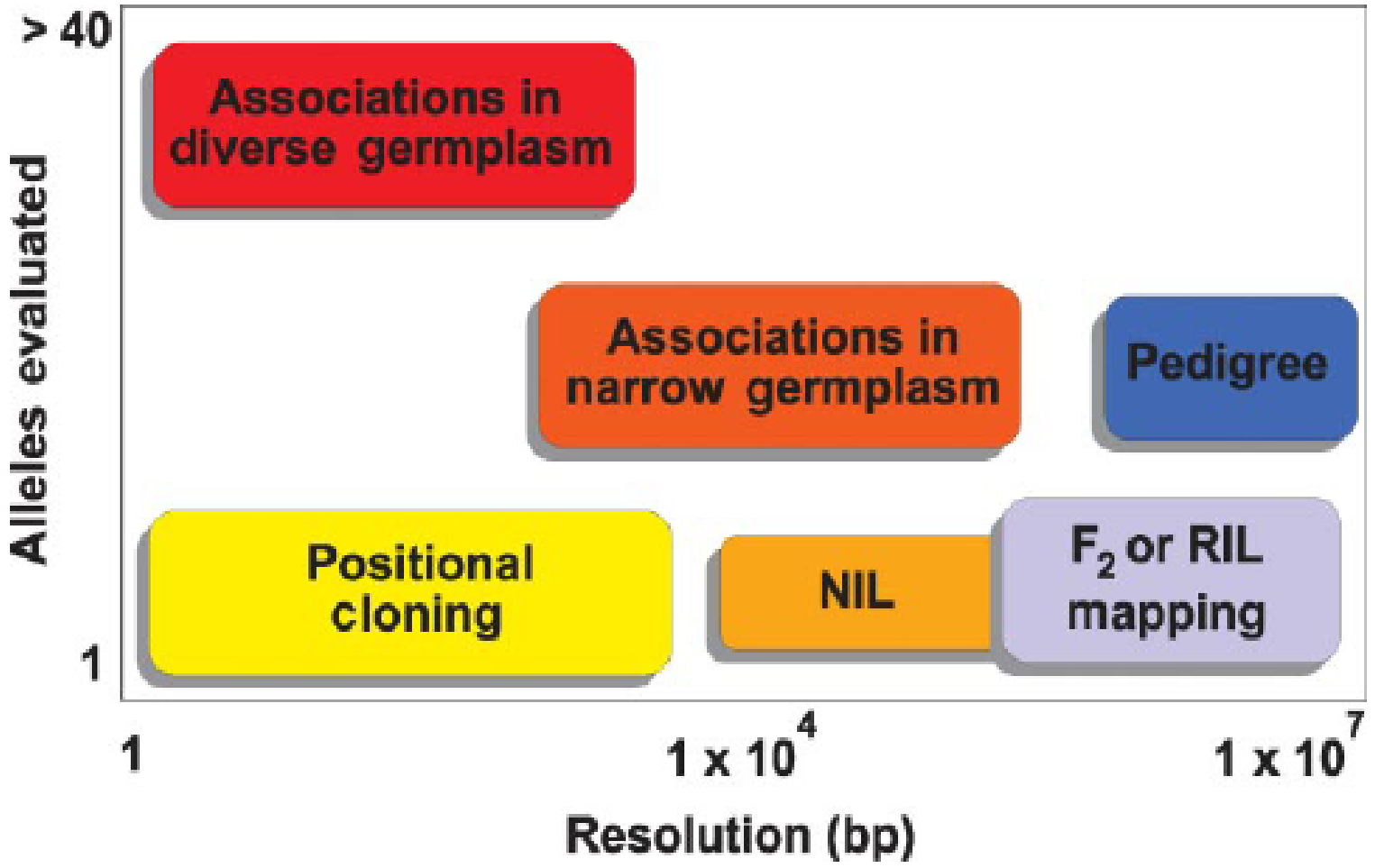
Yu et al. (2008) Genetics 178: 539–551.

Genotyping B73-rare SNPs to track recent recombination



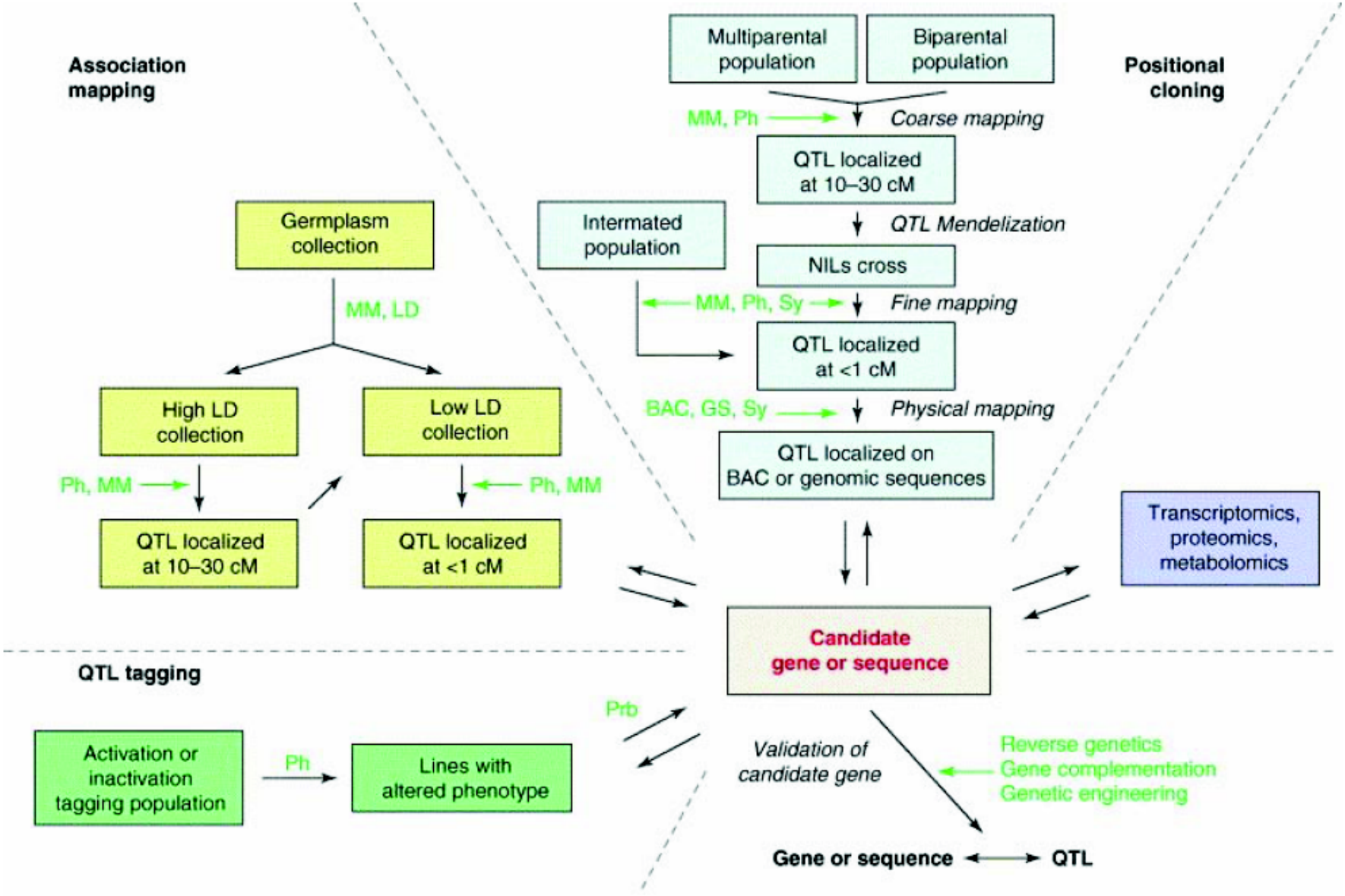
(a) Genotyping of both founders and RILs with CPS (common-parent-specific) markers (i.e., markers for which B73 has a rare allele) to track the inheritance of chromosome segments that resulted from recent recombination during RIL development tion during RIL development;
(b) genotyping of founders with high-density SNPs, projecting sequence polymorphism information (biallelic) from founders to RILs, and mapping in high resolution

Comparing various types of analyses for range of alleles vs.resolution



Association analysis of diverse germplasm are perfectly suited for sampling a wide range of alleles of high resolution - QTL!

Association genetics : challenging projects



An improved genome release (version Mt4.0) for the model legume Medicago truncatula

- *de novo* whole genome shotgun assembly, Illumina and 454 reads
- using optical and GBS genetic maps

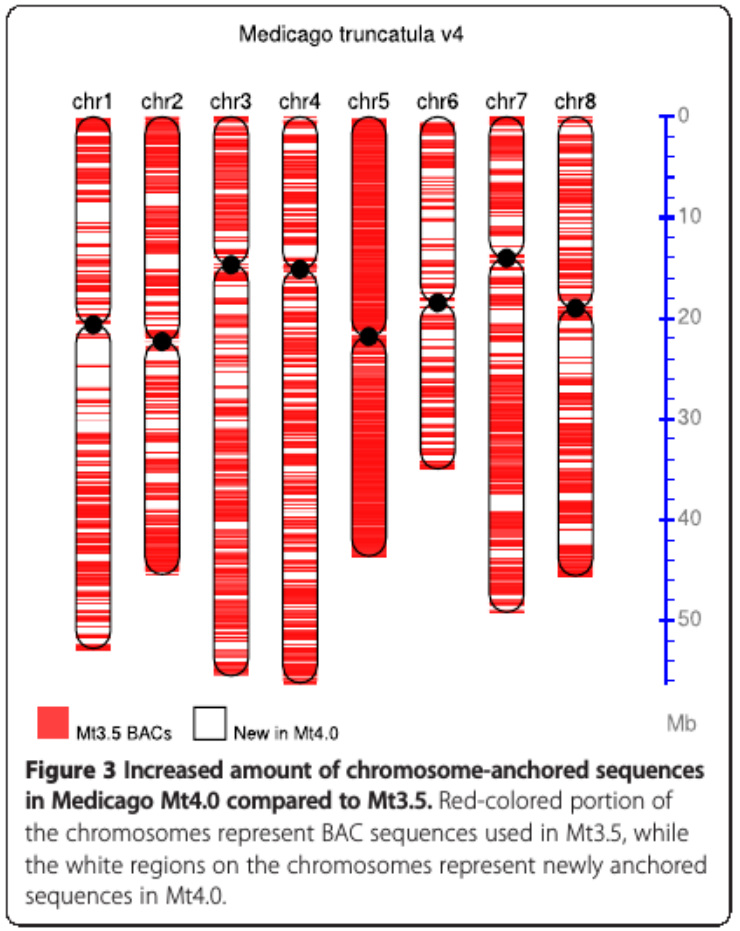


Figure 3 Increased amount of chromosome-anchored sequences in Medicago Mt4.0 compared to Mt3.5. Red-colored portion of the chromosomes represent BAC sequences used in Mt3.5, while the white regions on the chromosomes represent newly anchored sequences in Mt4.0.

Table 4 Characteristics of high confidence and low confidence gene sets

	High confidence (HC)	Low confidence (LC)
Number of genes	31,661	19,233
Number of single-exon genes	6,103 (19%)	5,351 (28%)
Number of multi-exon genes	25,558 (81%)	13,882 (72%)
Number of genes with alternative transcript variants	6,041 (19%)	347 (2%)
Number of predicted transcripts	42,481	19,838
Number of distinct exons	174,533	52,850
Mean gene locus size (first to last exon)	3,280	1,526
Mean transcript size (UTR, CDS)	1,618	841
Mean number of transcripts per gene	1.3	1.0
Mean number of distinct exons per gene	5.5	2.7
Mean exon size	308	296

Tang, H., Krishnakumar, V., Bidwell, S., Rosen, B., Chan, A., Zhou, S., Gentzbittel, L. Childs, K.L., Yandell, M., Gundlach, H., Mayer, K.F., Schwartz, D.C., Town, C.D., 2014. An improved genome release (version Mt4.0) for the model legume Medicago truncatula. BMC Genomics 15, 312.

Whole-genome nucleotide diversity, recombination, and linkage disequilibrium in the model legume *Medicago truncatula*

Antoine Branca[a,1], Timothy D. Paape[a], Peng Zhou[b], Roman Briskine[c], Andrew D. Farmer[d], Joann Mudge[d], Arvind K. Bharti[d], Jimmy E. Woodward[d], Gregory D. May[d], Laurent Gentzbittel[e], Cécile Ben[e], Roxanne Denny[b], Michael J. Sadowsky[f], Joëlle Ronfort[g], Thomas Bataillon[h], Nevin D. Young[a,b], and Peter Tiffin[a,2]

[a]Department of Plant Biology, University of Minnesota, Saint Paul, MN 55108; [b]Department of Plant Pathology, University of Minnesota, Saint Paul, MN 55108; [c]Department of Computer Science and Engineering, University of Minnesota, Minnepolis, MN 55455; [d]National Center for Genome Resources, Santa Fe, NM 87505; [e]Université de Toulouse, Laboratoire Symbiose et Pathologie des Plantes, ENSAT, 31326 Castanet-Tolosan, France; [f]Department of Soil, Water, and Climate, University of Minnesota, Saint Paul, MN 55108; [g]Unité Mixte de Recherche AGAP, Institut National de la Recherche Agronomique Montpellier domaine de Melgueil, F-34130 Mauguio, France; and [h]Bioinformatics Research Center and Institute of Biology, Aarhus University, 8000C Aarhus, Denmark

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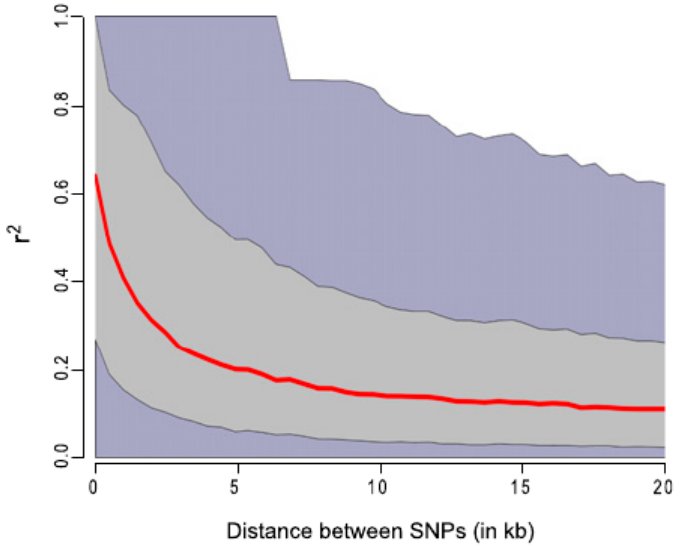


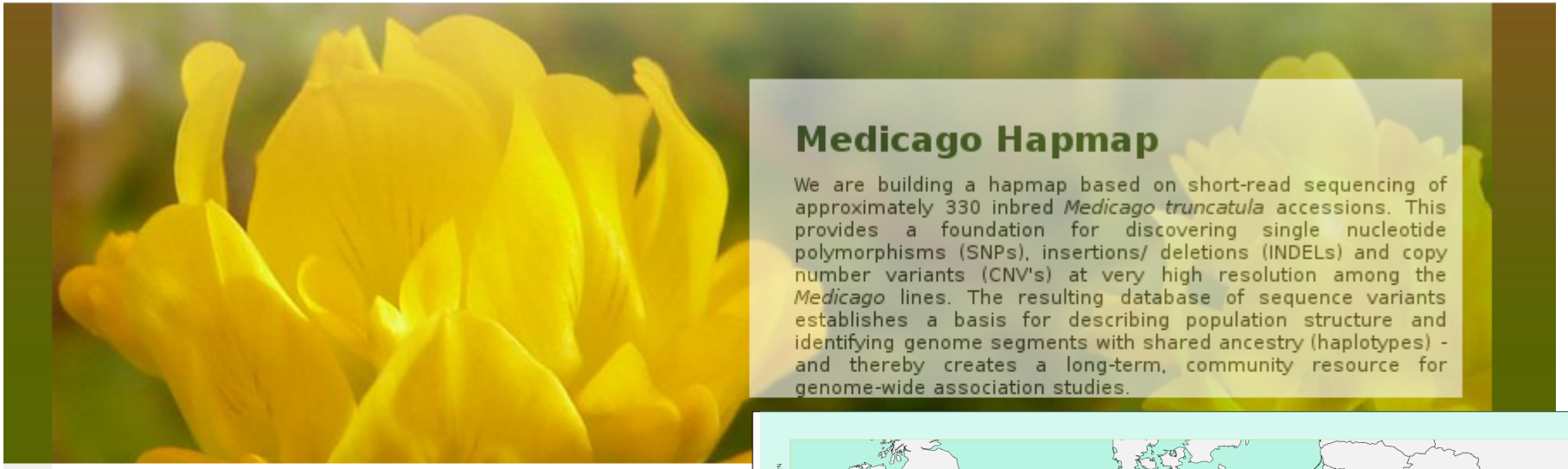
Fig. 6. Mean LD decay (red line) as measured by pairwise r^2, with the 50% and 90% ranges of values shown in light and dark gray, respectively.

NSF HapMap project

Medicago truncatula

HAPMAP PROJECT

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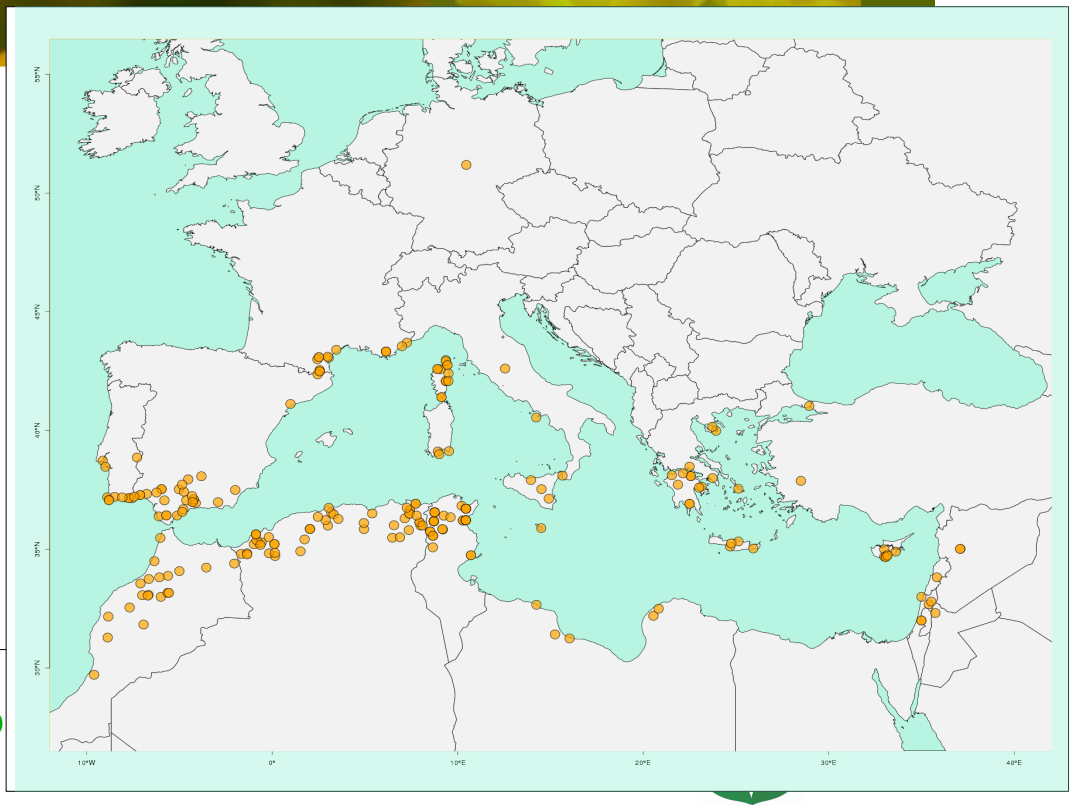
Medicago Hapmap

We are building a hapmap based on short-read sequencing of approximately 330 inbred *Medicago truncatula* accessions. This provides a foundation for discovering single nucleotide polymorphisms (SNPs), insertions/ deletions (INDELs) and copy number variants (CNV's) at very high resolution among the *Medicago* lines. The resulting database of sequence variants establishes a basis for describing population structure and identifying genome segments with shared ancestry (haplotypes) - and thereby creates a long-term, community resource for genome-wide association studies.

About this Project

We are developing a *Medicago* Hapmap as part of an international consortium consisting of the University of Minnesota, the National Center for Genome Resources (NCGR), Boyce Thompson Institute (BTI), J. Craig Venter Institute (JCVI) Hamline University, the University of Southern California, INRA-Montpellier, ENSAT-Toulouse, and the Noble Foundation.

Briefly, 384 inbred lines spanning the range of *Medicago* diversity are being resequenced using Illumina next generation technology. This provides a foundation for discovering single nucleotide polymorphisms (SNPs), insertions/deletions (INDELs) and copy number variants



INP ENSAT

Institut National Polytechnique, Ecole Nationale Supérieure Agronomique de Toulouse (INP-ENSAT)

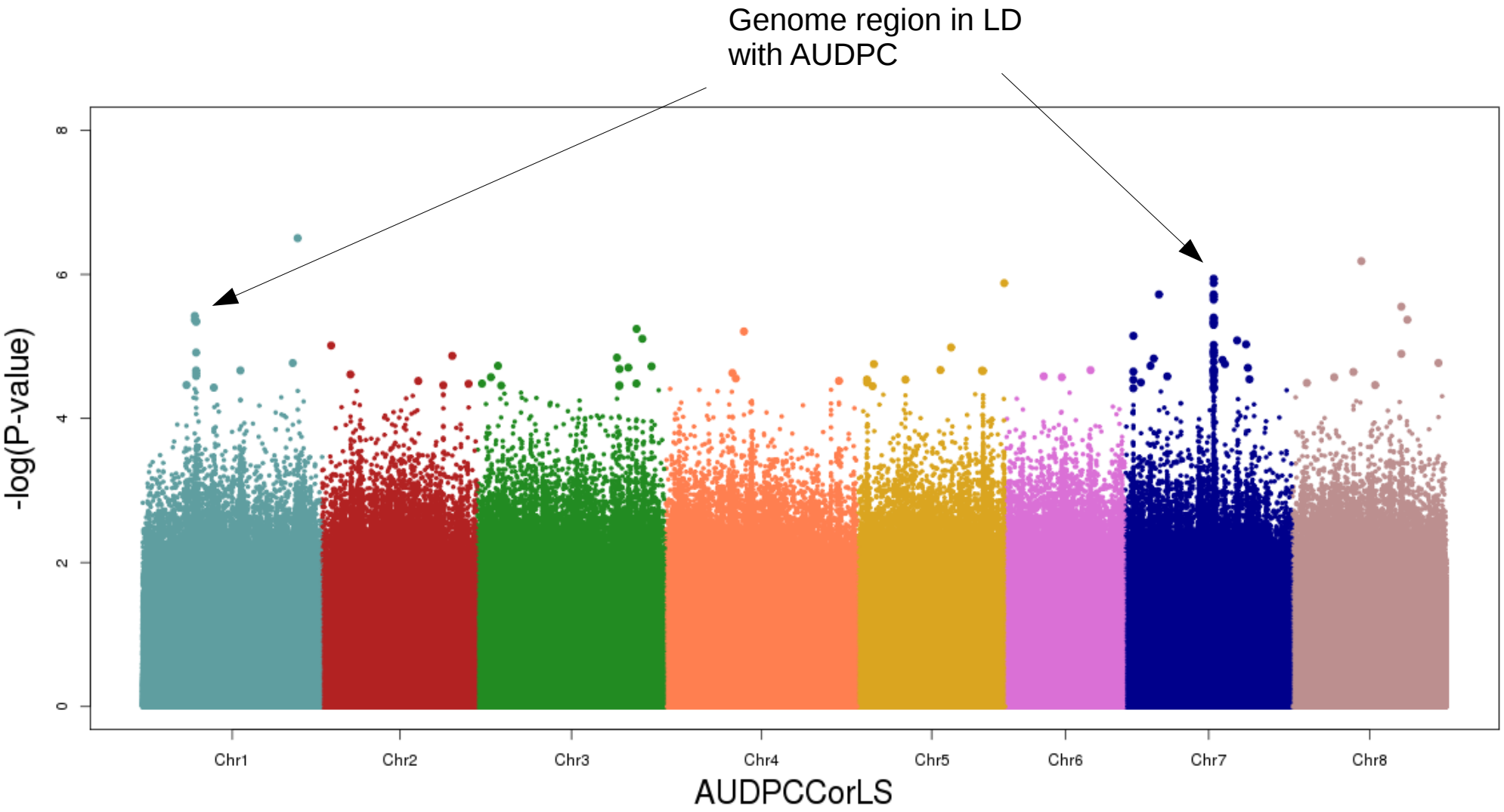
cnrs

EcoLab

Laboratoire écologie fonctionnelle et environnement

Internatio

GWAS for verticillium wilt



(Mazurier *et al. pending submission)*